Spitfire Pro

Ti:Sapphire Regenerative Amplifier Systems

Spitfire Pro F
Spitfire Pro 40F
Spitfire Pro PM

User's Manual

This laser product complies with performance standards of United States Code of Federal Regulations, Title 21, Chapter 1 - Food and Drug Administration, Department of Health and Human Services, Subchapter J - Parts 1040.10 (a), (1) or (2), as applicable.

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Preface

This manual contains the information required to safely operate and maintain the Spectra-Physics *Spitfire Pro* Ti:sapphire amplifier system. *Spitfire Pro* systems are available in several versions; this manual covers the *Spitfire Pro F*, *Spitfire Pro 40F*, and *Spitfire Pro PM* models.

Spitfire Pro systems amplify femtosecond (fs) pulses at near infrared and red wavelengths that are emitted by mode-locked Ti:sapphire lasers, such as those produced by the Spectra-Physics *Tsunami* or *Mai Tai*. Two standard repetition rates are available for amplified pulsed output, 1 kHz and 5 kHz, but the system can be adjusted for lower pulse repetition rates.

The system comprises three units: the *Spitfire Pro* head assembly, the *Timing Delay Generator* (*TDG*) control unit, and a laptop computer operating the *Spitfire* graphical control software. The *TDG* is a rack-mountable unit provided with all systems, as is the computer. The *Spitfire Pro* head assembly contains three internal assemblies: a pulse stretcher, a Ti:sapphire regenerative amplifier and a pulse compressor.

The stretcher and compressor are designed specifically for the pulse width of the input and output pulses for each *Spitfire Pro* model. The optics, including the pulse stretcher and compressor, are optimized for wavelength range, pulse width and repetition rate. This manual contains information on the optics and the stretcher and compressor configurations for each system.

Please note that *Spitfire Pro* specifications are guaranteed only when operated with a Spectra-Physics mode-locked Ti:sapphire seed laser and Spectra-Physics Q-switched pump laser, and which, in turn, are operated within the specifications described in this manual. The amplifier is designed specifically for the Spectra-Physics *Tsunami* or *Mai Tai* seed lasers and the *Empower* pump laser.

The first chapter of this manual, “Introduction,” contains a brief description of the *Spitfire Pro* systems.

Next is an important chapter on laser safety. The *Spitfire Pro* is a Class IV laser and, as such, emits laser radiation which can permanently damage eyes and skin. This chapter contains information about these hazards and offers suggestions on how to safeguard against them.

“General Description” contains an introductory description of laser theory, pulse stretching, laser amplification and pulse compression. Specifications for the various *Spitfire Pro* systems are included at the end of this chapter.

The next chapter is an overview of the external controls and external adjustments of the system. Please familiarize yourself with this material before operating the amplifier.

The following chapter describes the preparation needed to install the *Spitfire Pro* system. This chapter is only a guide to preparing the site for the initial set-up of the *Spitfire Pro* system. Please wait for the Spectra-Physics service engineer to install the system as part of your purchase agreement. Only personnel authorized by Spectra-Physics can install and set up your *Spitfire Pro* system.

"Operation" describes the routine operation of the *Spitfire Pro*. It is followed by a chapter that provides a detailed description of the beam path and internal adjustments of the amplifier. This information is needed should it become necessary to perform a simple re-alignment of the system using the procedures described later in this manual. A full alignment of the system should only be performed by an authorized Spectra-Physics representative.

The "Maintenance and Troubleshooting" section contains maintenance procedures that will allow you to keep your *Spitfire Pro* operational on a day-to-day basis. It also contains procedures you can perform to remedy any minor problems that might be encountered. The troubleshooting procedures can help you guide your Spectra-Physics field service engineer to the source of any major problems. *Do not attempt repairs yourself while the unit is still under warranty*; instead, report all problems to Spectra-Physics for warranty repair. This section includes a replacement parts list plus a list of world-wide Spectra-Physics service centers you can call if you need help.

This product has been tested and found to conform to "Directive 89/336/EEC for Electromagnetic Compatibility." Class A compliance was demonstrated for "EN 50081-2:1993 Emissions" and "EN 50082-1:1992 Immunity" as listed in the official Journal of the European Communities. Refer to the "CE Declaration of Conformity" on page 2-7.

Every effort has been made to ensure that the information in this manual is accurate. All information in this document is subject to change without notice. Spectra-Physics makes no representation or warranty, either express or implied with respect to this document. In no event will Spectra-Physics be liable for any direct, indirect, special, incidental or consequential damages resulting from any defects in this documentation. If you encounter any difficulty with the content or style of this manual, please let us know. The last page is a form to aid in bringing such problems to our attention.

Thank you for your purchase of Spectra-Physics instruments.

Environmental Specifications

CE Electrical Equipment Requirements

For information regarding the equipment needed to provide the electrical service listed under "Required Utilities" on page 5-3, please refer to specification EN-309, "Plug, Outlet and Socket Couplers for Industrial Uses," listed in the official *Journal of the European Communities.*

Environmental Specifications

The environmental conditions under which the laser system will function are listed below:

For indoor use only.

Altitude:	up to 2000 m
Temperatures:	10° C to 40° C
Maximum relative humidity:	80% non-condensing for temperatures up to 31° C.
Mains supply voltage:	do not exceed ±10% of the nominal voltage
Insulation category:	II
Pollution degree:	2

FCC Regulations

This equipment has been tested and found to comply with the limits for a Class A digital device pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with this instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference, in which case the user will be required to correct the interference at his own expense.

Modifications to the laser system not expressly approved by Spectra-Physics could void your right to operate the equipment.

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Warning Conventions

The following warnings are used throughout this manual to draw your attention to situations or procedures that require extra attention. They warn of hazards to your health, damage to equipment, sensitive procedures, and exceptional circumstances. All messages are set apart by a thin line above and below the text as shown here.



Laser radiation is present.



Condition or action may present a hazard to personal safety.



Condition or action may present an electrical hazard to personal safety.



Condition or action may cause damage to equipment.



Action may cause electrostatic discharge and cause damage to equipment.



Condition or action may cause poor performance or error.



Text describes exceptional circumstances or makes a special reference.



Do not touch.



Appropriate laser safety eyewear should be worn during this operation.



Refer to the manual before operating or using this device.

Standard Units

The following units, abbreviations, and prefixes are used in this Spectra-Physics manual:

Quantity	Unit	Abbreviation
mass	kilogram	kg
length	meter	m
time	second	s
frequency	Hertz	Hz
force	Newton	N
energy	Joule	J
power	Watt	W
electric current	Ampere	A
electric charge	Coulomb	C
electric potential	volt	V
resistance	Ohm	Ω
inductance	Henry	H
magnetic flux	Weber	Wb
magnetic flux density	Tesla	T
luminous intensity	candela	cd
temperature	Celsius	C
pressure	Pascal	Pa
capacitance	Farad	F
angle	radian	rad

Prefixes								
tera	(10^{12})	T	deci	(10^{-1})	d	nano	(10^{-9})	n
giga	(10^{9})	G	centi	(10^{-2})	c	pico	(10^{-12})	p
mega	(10^{6})	M	mill	(10^{-3})	m	femto	(10^{-15})	f
kilo	(10^{3})	k	micro	(10^{-6})	μ	atto	(10^{-18})	a

Abbreviations

The following is a list of abbreviations used in Spectra-Physics manuals:

ac	alternating current
AOM	acousto-optic modulator
APM	active pulse mode locking
AR	anti reflection
bi-fi	birefringent filter
CDRH	Center of Devices and Radiological Health
CE	European Union
CPA	chirpped pulse applification
CPM	colliding pulse mode locking
CW	continuous wave
dc	direct current
E/O	electro-optic
fs	femtosecond or 10^{-15} second
GTI	Gires-Toutnois Interferometer
GVD	group velocity dispersion
HR	high reflector
IR	infrared
OC	output coupler
OD	outside dimension
ps	picosecond or 10^{-12} second
PZT	piezo-electric transducer
RF	radio frequency
SBR	saturable Bragg reflector
SCFH	standard cubic feet per hour
SPM	self phase modulation
TEM	transverse electromagnetic mode
Ti:sapphire	Titanium-doped Sapphire
UV	ultraviolet
λ	wavelength

Unpacking and Inspection

Unpacking Your System

Your *Spitfire Pro* laser amplifier was packed with great care, and the containers were inspected prior to shipment. Upon receiving the system, immediately inspect the outside of the shipping containers. If there is any major damage (holes in the containers, crushing, etc.), insist that a representative of the carrier be present when you unpack the contents.

It is important that these instructions are followed carefully. The system was precisely aligned at the factory, then packed and shipped in a manner to preserve that alignment. Handle the system with care while unpacking to preserve this condition.

Carefully inspect the laser system as you unpack it. If any damage is evident, such as dents or scratches on the covers or broken parts, etc., immediately notify the carrier and your Spectra-Physics sales representative.

Keep the shipping containers. If you file a damage claim, they may be needed to demonstrate that the damage occurred as a result of shipping. If the system is ever returned for service, the specially designed containers assure adequate protection.

Warning! 

Spectra-Physics is responsible for the safety, reliability and performance of the *Spitfire Pro* amplifier only under the following conditions:

- All field-installable options, modifications or repairs are performed by persons trained and authorized by Spectra-Physics.
- The equipment is used in accordance with the instructions provided in this manual.

System Components

The system is shipped in two separate containers:

- one contains the *Spitfire Pro* assembly
- one contains the rack-mountable *TDG* controller and accessory kit (see below) and the system laptop computer

Also included, if ordered, are the Spectra-Physics *Empower* pump laser and a Spectra-Physics *Tsunami* or *Mai Tai* seed laser system.

Accessory Kit

Included with the laser system is this manual, a CD holding the *Spitfire Pro* software, a packing slip listing all the components shipped with this order, and an accessory kit containing the following items:

- (4) chassis clamps
- (4) cables
- (4) ¼–20 screws
- knob
- adjustable iris aperture, and base
- 20 ft (6 m) of ¼ in. OD tubing

Chapter 1 Introduction

The Spectra-Physics *Spitfire Pro* systems amplify individual laser pulses selected from a stream of pulses that are produced by a separate, mode-locked Ti:sapphire laser. An input pulse with an energy of only a few nanojoules may be amplified to an energy exceeding 1 millijoule. Specific *Spitfire Pro* models can produce amplified pulses ranging in duration from less than 40 femtoseconds up to about 80 picoseconds.

The maximum output energy of a solid-state amplifier is normally limited by the optical damage threshold of the crystalline material used in the system. The *Spitfire Pro* regenerative amplifier circumvents this limitation by means of "chirped pulse amplification" (CPA). This technique, originally developed for radar systems, first temporally stretches a pulse to reduce its peak power, then amplifies it, and finally recompresses the pulse to a width close to its original duration. This results in greatly increased peak power while avoiding optical damage to the amplifier.

The Spitfire Pro System

The *Spitfire Pro* amplifier itself comprises two main components:

- the *Spitfire Pro* amplifier head assembly
- the Timing and Delay Generator (*TDG*)

A complete system requires a pump laser to energize the *Spitfire Pro* amplifier and a seed laser to provide the input pulses. Figure 1-1 shows a typical application: a *Spitfire Pro* pumps an optical parametric amplifier, and is seeded by a *Tsunami* laser system and pumped by an *Empower* laser.

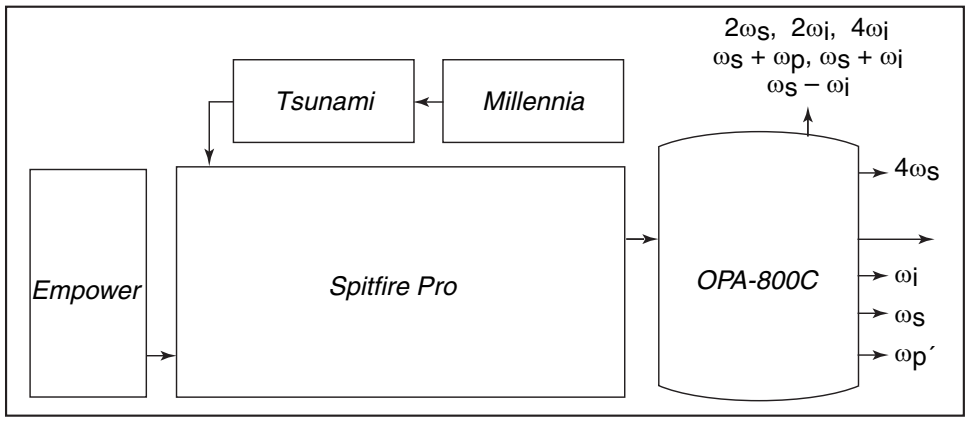


Figure 1-1: A typical layout showing the *Spitfire Pro* pumping a Spectra-Physics *OPA-800C*

Configurations

The *Spitfire Pro* system is available in a variety of models that produce a wide range of pulse wavelengths, energies and durations. Table 1-1 lists the models covered by this manual.

Each *Spitfire Pro* model can be ordered preset to either a 1 kHz or 5 kHz pulse repetition rate, which then may be reduced to a lower rate by the user. Some systems may also be converted from one model to another. Contact your Spectra-Physics representative for more information about these options.

Table 1-1: *Spitfire Pro* Configuration Matrix

Amplifier Model	Pulse Width	Description
Spitfire Pro F	<130 fs	"standard" version
Spitfire Pro 40F	< 40 fs	ultra-short output pulses
Spitfire Pro PM	<2 ps	"pico-mask" version – produces picosecond pulses using a femtosecond seed laser

See Table 3-1 for complete specifications for the various systems. The configurations listed in Table 1-1 are also available in high-power models, the *Spitfire Pro HP* systems. If interested, contact your Spectra-Physics representative for more information about *Spitfire Pro HP* systems.

The versions of the *Spitfire Pro* listed in Table 1-1, with the exception of the *Spitfire Pro 40F*, are available in three standard wavelength ranges as determined by the optics set used. The *Spitfire Pro 40F* is available only with Optics Set 1.

Table 1-2: *Spitfire Pro* Optics Sets

Optics Set	Output Wavelength Range
Optics Set 1	750 nm – 840 nm
Optics Set 2	840 nm – 870 nm
Optics Set 3	870 nm – 900 nm

Each optics sets comes in two versions, depending on whether the system will be run at 1 kHz or 5 kHz. The *Spitfire Pro* system is delivered with the optics set(s) appropriate for the wavelength range and repetition rate that was specified when the system was ordered.

Custom Spitfire Pro Systems

Custom versions of the *Spitfire Pro* are available that produce pulses at wavelengths, repetition rates and energy levels different than those listed in Table 1-1. Again, contact your Spectra-Physics representative for more information.

The Spitfire Pro Amplifier

The *Spitfire Pro* amplifier head contains the optics and optomechanical devices for stretching, selecting, amplifying and compressing pulses from a seed laser (such as a Spectra-Physics *Mai Tai* or *Tsunami*). The *Spitfire Pro* amplifier head contains the following three internal sub-assemblies:

- the optical pulse stretcher
- the regenerative amplifier
- the optical pulse compressor

These sub-assemblies are each carefully optimized for the chosen wavelength range, the repetition rate of the amplified output, and the duration of the amplified pulses.

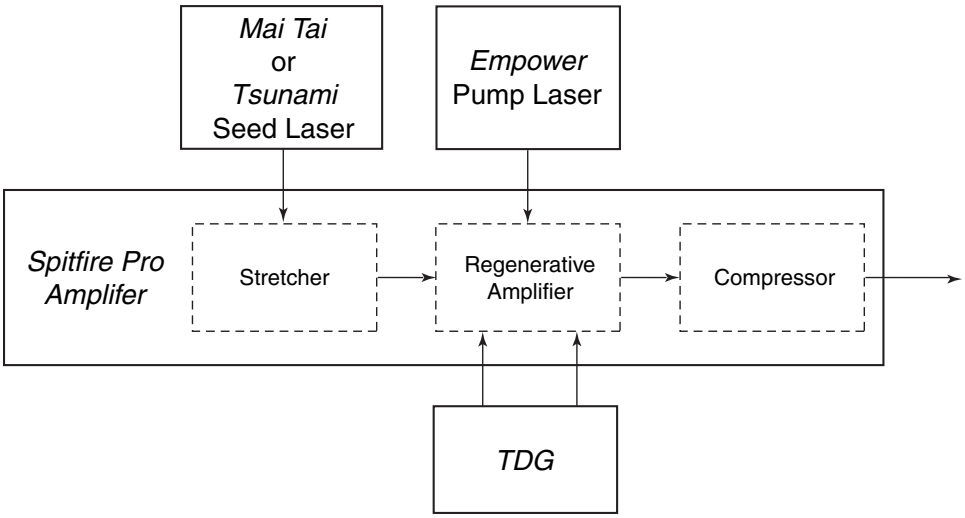


Figure 1-2: Block Diagram for the *Spitfire Pro F, 40F* and *PM*

Incoming seed pulses are stretched using a multi-pass grating and mirror combination. The *TDG* provides the synchronization and control needed to select individual pulses from the train of stretched seed pulses and capture them in the regenerative amplifier. The selected, stretched pulses then make multiple passes through the regenerative amplifier, increasing in energy.

After the pulses are amplified, the *TDG* provides the timing control to direct the amplified pulses into the compressor where they are shortened close to their original duration using a second grating/mirror combination. The pulses are then directed out of the *Spitfire Pro*.

Titanium Sapphire

Spitfire Pro amplification is provided by a titanium-doped sapphire (Ti:sapphire) crystal. Ti:sapphire was selected because of two important properties: (a) it has a broad absorption band in the blue and green wavelengths that allows it to be pumped by the frequency-doubled output of a Nd:YLF laser, and (b) it is tunable over a broad emission band of wavelengths in the near infrared. For a more detailed explanation of the theory of operation of the *Spitfire Pro*, refer to Chapter 3, “General Description.”

Chapter 2 Laser Safety

Danger!





The Spectra-Physics *Spitfire Pro*® amplifier is classified as a C*lass IV—High Power Laser* whose beam is, by definition, a safety and fire hazard. Take precautions to prevent accidental exposure to both direct and reflected beams. Diffuse as well as specular beam reflections can cause severe eye or skin damage.

Because the output wavelength is typically between 700 and 1000 nm (from red to infrared), the *Spitfire Pro* output beam is often invisible and therefore especially dangerous. This type of infrared radiation passes easily through the cornea of the eye, and, when focused on the retina, can cause instantaneous and permanent damage.

Precautions For The Safe Operation of Class IV High Power Lasers

- Wear protective eyewear at all times; selection depends on the wavelength and intensity of the radiation, the conditions of use, and the visual function required. Protective eyewear is available from suppliers listed in the *Laser Focus World*, *Lasers and Optronics*, and *Photonics Spectra* buyer's guides. Consult the ANSI and ACGIH standards listed at the end of this section for guidance.
- Maintain a high ambient light level in the laser operation area so the eye's pupil remains constricted, reducing the possibility of damage.
- Avoid looking at the output beam; even diffuse reflections are hazardous.
- Avoid blocking the output beam or its reflections with any part of the body.
- Establish a controlled access area for laser operation. Limit access to those trained in the principles of laser safety.
- Enclose beam paths wherever possible.
- Post prominent warning signs near the laser operating area (Figure 2-1).
- Set up experiments so the laser beam is either above or below eye level.
- Set up shields to prevent any unnecessary specular reflections or beams from escaping the laser operation area.
- Set up a beam dump to capture the laser beam and prevent accidental exposure (Figure 2-2).

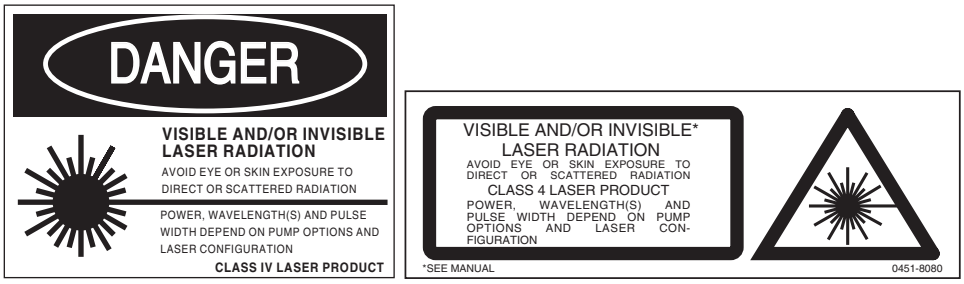


Figure 2-1: These CE and CDRH standard safety warning labels would be appropriate for use as entry warning signs (EN 60825-1, ANSI Z136.1 Section 4.7).

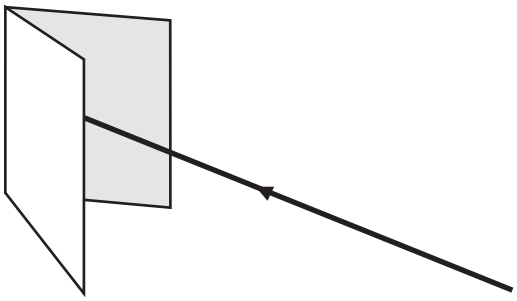


Figure 2-2: Folded Metal Beam Target

Caution! 

Use of controls or adjustments, or the performance of procedures other than those specified herein may result in hazardous radiation exposure.

Follow the instructions contained in this manual for safe operation of your laser. At all times during operation, maintenance, or service of your laser, avoid unnecessary exposure to laser or collateral radiation* that exceeds the acceptable emission limits listed in "Performance Standards for Laser Products," *United States Code of Federal Regulations*, 21CFR1040 10(d).

Safety Devices

The *Spitfire Pro* cannot generate output energy without being pumped and seeded by other lasers. When either the pump laser beam or the seed laser beam is blocked, the *Spitfire Pro* will produce no amplified output. In this sense, the safety devices of the seed and pump lasers act as safety devices for the *Spitfire Pro*. (Note that in some circumstances, even though the *Spitfire Pro* system is inactive, a portion of the unamplified seed laser beam can pass through the *Spitfire Pro* and exit the output port.)

Removing the cover of the *Spitfire Pro* will activate shutters that block both the seed beam and the pump beam from entering the amplifier.

* *Any electronic product radiation, except laser radiation, emitted by a laser product as a result of, or necessary for, the operation of a laser incorporated into that product.*

Fuses

The *Spitfire Pro Timing and Delay Generator (TDG*) uses one of the following fuses, as appropriate for the local line voltage:

120 Vac	220 Vac
6 A 250 V, Slow Blow	3 A 250 V, Slow Blow

Maximum Emission Levels

The following is the maximum emission level possible for the *Spitfire Pro* amplifier. Use this information for selecting appropriate laser safety eyewear and implementing appropriate safety procedures. This value does not imply actual system power or specifications.

Emission Wavelength	Maximum Power
690 to 1080 nm	10 W

CDRH Compliance

This laser product complies with Title 21 of the *United States Code of Federal Regulations*, Chapter 1, subchapter J, parts 1040.10 and 1040.11, as applicable. To maintain compliance with these regulations, once a year, or whenever the product has been subjected to adverse environmental conditions (e.g., fire, flood, mechanical shock, spilled solvent, etc.), check to see that all features of the product identified on the CDRH Radiation Control Drawing (found later in this chapter) function properly. Also, make sure that all warning labels remain firmly attached.

CDRH Requirements for Operating the Spitfire Pro Using PC Control

The *Spitfire Pro* system complies with all CDRH safety standards when operated using the *Spitfire* GUI interface installed on the personal computer. The following features are provided to satisfy CDRH regulation requirements:

- **An emission indicator—**that indicates laser energy is present, imminent or can be accessed. This indicator is present on all *Spitfire* GUI displays that affect the output of the laser.
- **A safety key—**to prevent unauthorized use. In addition to the power supply interlock keys, the password feature of the personal computer (in the CMOS Setup program) or the Windows® operating system is available as a safety key for this requirement

Windows is a registered trademark of the Microsoft Corporation.

CDRH Radiation Control Drawings

Refer to the warning labels in Figure 2-4.

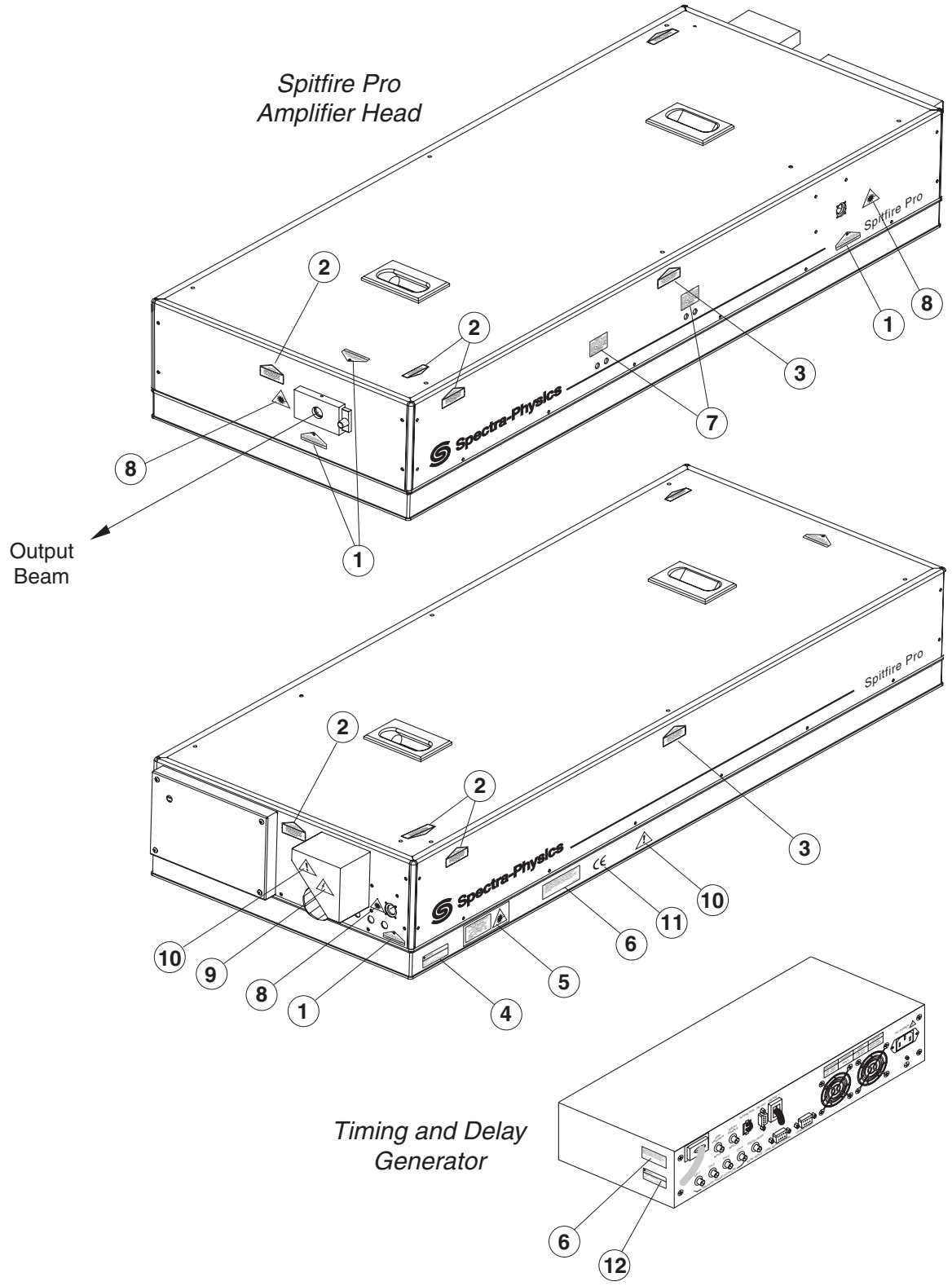
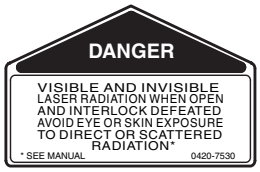


Figure 2-3: CDRH Radiation Control Drawing

CE/CDRH Warning Labels



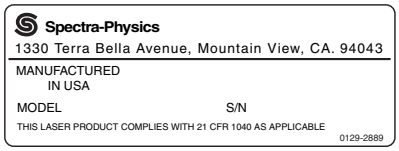
Aperture Label, Laser Head (1)



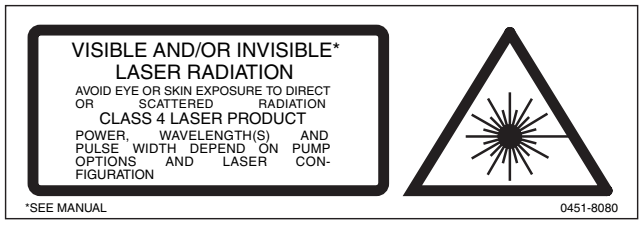
Interlocked Cover Warning Label (2)



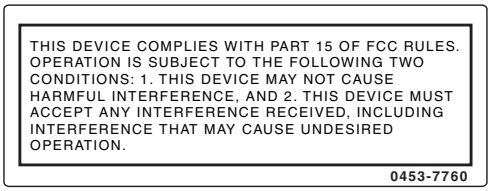
EMI Interlock Caution Label (3)



Identification/Certification Label (4)



Class 4 Laser Warning Label (5)



FCC Compliance Label (6)



Danger Radiation Label (7)



CE Aperture Label (8)



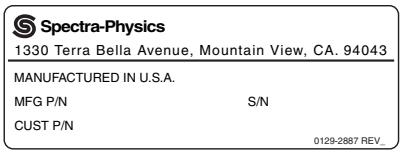
Electrical Hazzard Label (9)



CE Caution Label (10)



CE Certification Label (11)



Model/Serial Identification Label (12)

Figure 2-4: CDRH Warning Labels

Label Translations

For safety, the following translations are provided for non-English speaking personnel. The number in parenthesis in the first column corresponds to the label number listed on the previous page.

Table 2-1: Label Translations

Label No.	French	German	Spanish	Dutch
Aperture Label (1)	Ouverture Laser - Exposition Dangereuse - Un Rayonnement laser visible et invisible est émis par cette ouverture.	Austritt von sichtbarer und unsichtbarer Laserstrahlung! Bestrahlung vermeiden!	Por esta abertura se emite radiación láser visible e invisible; evite la exposición.	Vanuit dit apertuur wordt zichtbare en onzichtbare laserstraling geemitteerd! Vermijd blootstelling!
Danger, Defeatable Interlock (2)	Attention- Rayonnement Laser visible et invisible en cas D'Ouverture et lorsque la sécurité est neutralisée; exposition dangereuse de l'oeil ou de la peau au rayonnement direct ou diffus.	Gefahr! Austritt von sichtbarer und unsichtbarer Laserstrahlung, wenn Abdeckung geöffnet und Sicherheitsverschluß überbrückt! Bestrahlung von Auge oder Haut durch direkte oder Streustrahlung vermeiden! Bedienungsanleitung beachten!	Peligro, al abrir y retirar el dispositivo de seguridad existe radiación láser visible e invisible; evite que los ojos o la piel queden expuestos tanto a la radiación directa como a la dispersa.	Gevaar! Zichtbare en onzichtbare laserstraling wanneer geopend en interlock uitgeschakeld! Vermijd blootstelling van oog of huid aan directe straling of terugkaatsingen daarvan!
Caution, Defeatable Interlock (EMI) (3)	Attention. Rayonnement visible et invisible dangereux en cas d'ouverture et lorsque la sécurité est neutralisée.	Vorsicht! Austritt von sichtbarer und unsichtbarer schädlicher elektromagnetischer Strahlung, wenn Abdeckung geöffnet und Sicherheitsverschluß überbrückt. Bedienungsanleitung beachten!	Precaución, radiación peligrosa electromagnética visible e invisible con el dispositivo de seguridad abierto o con su indicación alterada.	Let op! Zichtbare en onzichtbare gevaarlijke elektromagnetische straling wanneer geopend en interlock uitgeschakeld!
Class IV Laser Warning Label (5)	Rayonnement Laser Visible et Invisible en Cas D'Ouverture et lorsque la sécurité est neutralisée; exposition dangereuse de l'oeil ou de la peau au rayonnement direct ou diffus. Laser de Classe 4. Puissance et longueurs D'onde dépendant de la configuration et de la puissance de pompe. EN60825-1:1994	Gefahr! Sichtbare und unsichtbare Laserstrahlung! Bestrahlung von Auge oder Haut durch direkte oder Streustrahlung vermeiden! Laserklasse IV. Leistung, Wellenlänge und Pulsbreite sind abhängig von Pumpquelle und Laserkonfiguration. Bedienungsanleitung beachten! EN60825-1:1994	Al abrir y retirar el dispositivo de seguridad existe radiación láser visible e invisible; evite que los ojos o la piel queden expuestos tanto a la radiación directa como a la dispersa. Producto láser clase 4. Potencia, longitud de onda y anchura de pulso dependen de las opciones de bombeo y de la configuración del láser. EN60825-1:1994	Gevaar! Zichtbare en onzichtbare laserstraling! Vermijd blootstelling van oog of huid aan directe straling of terugkaatsingen daarvan! Klas IV laser produkt. Vermogen, golflengte en pulsduur afhankelijk van pomp opties en laser konfiguratie. EN60825-1:1994
Danger Radiation Label (7)	DANGER! Emission laser lorsque les bouchons d'accès aux ajustements sont retirés. Exposition dangereuse de l'oeil ou de la peau au rayonnement direct ou diffus.	GEFAHR! Austritt von Laserstrahlung wenn Justierzugang geoeffnet. Bestrahlung von Auge oder Haut durch direkte oder Streustrahlung vermeiden!	PELIGRO! Radiación de Láser esta presente cuando el tapón de ajusta miento no esta puesto. Evite que los ojos o la piel queden expuestos tanto a la radiación directa como a la dispersa.	GEVAAR! Laser-stralingsgevaar wanneer de doppen van de afregelpoorten verwijderd zijn. Vermijd blootstelling van oog of huid aan directe straling of terugkaatsingen daarvan!

CE Declaration of Conformity

We,

Spectra-Physics
1330 Terra Bella Avenue
Mountain View, CA. 94043
United States of America

declare under our sole responsibility that the

Spitfire Pro Multi-Kilohertz Ti:Sapphire Regenerative Amplifier

manufactured after December 14, 2004

meets the intent of "EMC Directive 89/336/EEC (2004/C 98/05) for Electromagnetic Compatibility" and "Directive 73/23/EEC (1973), the Low Voltage Directive." Compliance was demonstrated to the following Specifications as listed in the official *Journal of the European Communities*:

89/336/EEC: 2004/C 98/05, EMC Directive

- **EN 61000-6-4: 2001:** Emission standard for industrial environments — Generic Standards
- **EN 55011: 1998+A1+A2:** Industrial, scientific and medical (ISM) radio-frequency equipment radio disturbance characteristics
- **EN 61000-3-2: 2000:** Limits for harmonic current emissions (equipment input current up to and including16 A per phase)
- **EN 61000-3-3: 1995+A1 section 3:** Limitation of voltage changes, voltage fluctuation and flicker
- **EN 61000-6-2: 2001:** Generic Standards—Immunity for industrial environments
- **EN 61000-4-2: 1995+A1+A2, part 4 section 2:** Electrostatic discharge immunity test
- **EN 61000-4-3: 2002+A1 part 4 section 3:** Testing and measurement techniques—Radiated, radio frequency, electromagnetic field immunity test
- **EN 61000-4-4: 1995+A1+A2 part 4-4:** Testing and measurement techniques—Electrical fast transient/burst immunity test
- **EN 61000-4-5: 1995+A1:** Testing and measurement techniques—Surge immunity test
- **EN 61000-4-6: 1996+A1 part 4-6:** Testing and measurement techniques—Immunity to conducted disturbances, induced by radio frequency fields
- **EN 61000-4-8: 1993:** Testing and measurement techniques—Power, frequency, magnetic field immunity tests
- **EN 61000-4-11: 1994+A1:** Testing and measurement techniques—Voltage dips, short interruptions and voltage variations immunity tests

73/23/EEC: 1973, Low Voltage Directive

- **EN60825-1: Edition 1.2 (2001) part 1:** Equipment classification, requirements, and users guide, safety of laser products
- **EN60950: 2001 (2000 Ed. + Corr 1 + Corr 2):** Safety of Information Technology Equipment
- **EN 61010-1: (2001 + Corr 1 + Corr 2):** part 1: General requirements—Safety requirements for electrical equipment for measurement

I, the undersigned, hereby declare that the equipment specified above conforms to the above Directives and Standards.



Bruce Craig
Vice President
Spectra-Physics / Newport
December 14, 2004

Sources for Additional Information

Laser Safety Standards

Safe Use of Lasers (Z136.1)
American National Standards Institute (ANSI)
11 West 42nd Street
New York, NY 10036
Tel: (212) 642-4900

Occupational Safety and Health Administration (Publication 8.1-7)
U. S. Department of Labor
200 Constitution Avenue N. W., Room N3647
Washington, DC 20210
Tel: (202) 693-1999
Internet: www.osha.gov

A Guide for Control of Laser Hazards, 4th Edition, Publication #0165
American Conference of Governmental and
Industrial Hygienists (ACGIH)
1330 Kemper Meadow Drive
Cincinnati, OH 45240
Tel: (513) 742-2020
Internet: www.acgih.org/home.htm

Laser Institute of America
13501 Ingenuity Drive, Suite 128
Orlando, FL 32826
Tel: (800) 345-2737
Internet: www.laserinstitute.org

Compliance Engineering
Canon Communications LLC
11444 W. Olympic Boulevard
Los Angeles, CA 90064
Tel: (310) 445-4200

International Electrotechnical Commission
Journal of the European Communities
EN 60825-1 Safety of Laser Products — Part 1: Equipment classification, requirements and user's guide
Tel: +41 22-919-0211 Fax: +41 22-919-0300
Internet: www.iec.ch

Cenelec
35, Rue de Stassartstraat
B-1050 Brussels, Belgium
Tel: +32 2 519 68 71
Internet: www.cenelec.org

Document Center, Inc.
111 Industrial Road, Suite 9
Belmont, CA 94002
Tel: (650) 591-7600
Internet: www.document-center.com

Equipment and Training

Laser Safety Guide
Laser Institute of America
13501 Ingenuity Drive, Suite 128
Orlando, FL 32826
Tel: (800) 34LASER
Internet: www.laserinstitute.org

Laser Focus World Buyer's Guide
Laser Focus World
Pennwell Publishing
98 Spit Rock Road
Nashua, NH 03062
Tel: (603) 891-0123
Internet: lfw.pennnet.com/home.cfm

Photonics Spectra Buyer's Guide
Photonics Spectra
Laurin Publications
Berkshire Common
PO Box 4949
Pittsfield, MA 01202-4949
Tel: (413) 499-0514
Internet: www.photonics.com

Chapter 3 General Description

The *Spitfire Pro* amplifier system contains all the components necessary to amplify low-energy Ti:sapphire laser pulses to energy levels as high as a millijoule. The *Spitfire Pro* amplifier head assembly comprises the optical stretcher, the regenerative amplifier and the optical compressor. The femtosecond seed pulses that are amplified are provided by a separate mode-locked Ti:sapphire laser system.

The *Spitfire Pro* system also includes the *Timing and Delay Generator*, or *TDG*, which provides the precise timing required to select pulses for amplification and to eject them from the amplifier. The functions of both the amplifier and *TDG* are described in this chapter.

Ti:Sapphire

Ti:sapphire is a crystalline material produced by introducing Ti_2O_3 into a melt of Al_2O_3, where Ti^{3+} (titanium) ions replace a small percentage of the Al^{3+} (aluminium) ions. A boule of material is then grown from this melt. The Ti^{3+} ion is responsible for the lasing action in Ti:sapphire. The electronic ground state of the Ti^{3+} ion is split into a pair of vibrationally broadened levels as shown in Figure 3-1.

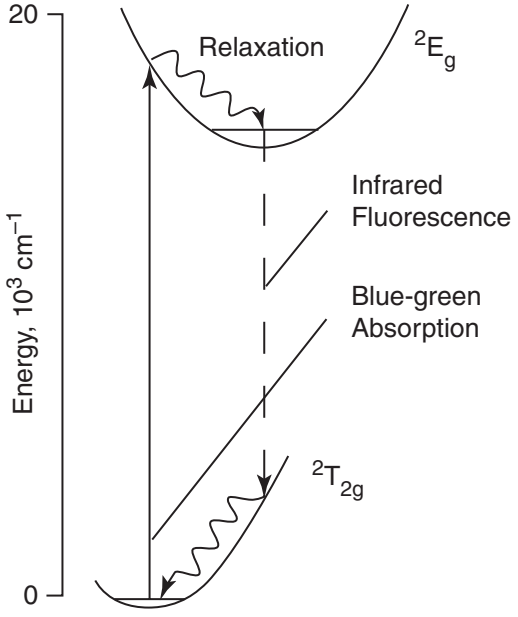


Figure 3-1: Energy Level Structure of Ti:Sapphire

Absorption transitions occur over a broad range of wavelengths from 400 to 600 nm (only one of which is shown in Figure 3-1). Fluorescence transitions occur from the lower vibrational levels of the excited state to the upper vibrational levels of the ground state. The resulting emission and absorption spectra are shown in Figure 3-2.

Although the fluorescence band extends from wavelengths as short as 600 nm to wavelengths greater than 1000 nm, lasing action is only possible at wavelengths longer than 670 nm because the long wavelength side of the absorption band overlaps the short wavelength end of the fluorescence spectrum. Additionally, the tuning range may be reduced by variations in mirror coatings, tuning element losses, pump power and pump mode quality.

Nevertheless, Ti:sapphire possesses the broadest continuous wavelength tuning range of any commercially available laser. As discussed in the following sections, this broad tuning range allows Ti:sapphire lasers to produce and amplify optical pulses of extremely short duration. As a corollary, the same factors that allow Ti:sapphire a broad, tunable wavelength range might also affect the production and amplification of these ultrashort pulses.

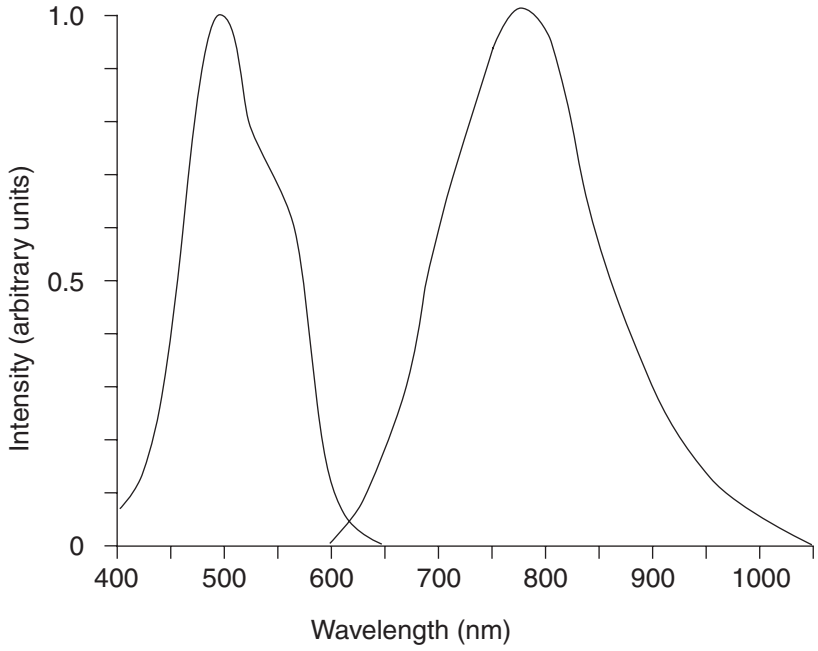


Figure 3-2: Absorption and Emission Spectra of Ti:Sapphire

The Ti:sapphire crystal is highly resistant to thermally induced stress. This resistance allows it to be optically pumped at relatively high average powers without danger of fracture. However, it cannot handle the high peak powers that would result from directly amplifying femtosecond pulses. A technique called Chirped Pulse Amplification, or CPA, which temporally stretches the pulse prior to amplification and then recompresses it following amplification, circumvents this limitation.

Chirped Pulse Amplification

When an intense beam travels through a Ti:sapphire crystal, it tends to "self-focus." Self-focusing is a nonlinear optical effect in which an intense light beam modifies the refractive index of the material it is passing through, causing the beam to focus and intensify even further. This can potentially result in a run-away condition that causes permanent damage to the crystal. Self-focusing makes it necessary to limit the peak power of a pulse in the Ti:sapphire crystal to less than 10 GW/cm^2.

Chirped Pulse Amplification (CPA) allows a Ti:sapphire crystal to be used to amplify pulses beyond this peak power, while keeping the power density in the amplifier below the damage threshold of the crystal. CPA is accomplished in three steps. The first step stretches the very short seed pulse that is supplied by a stable, mode-locked picosecond or femtosecond laser. Stretching the pulse (i.e., increasing its duration) reduces its peak power, which greatly reduces the probability of damage to the Ti:sapphire amplifier crystal.

The second step amplifies the stretched pulse: a pump laser provides a synchronous energy pulse to the Ti:sapphire crystal to excite it just prior to the arrival of the stretched seed pulse. The seed pulse causes stimulated emission, which amplifies the pulse at the same wavelength and direction. (This is in contrast to "spontaneous emission" within the gain medium that typically is amplified to become laser output in other systems.)

The third step recompresses the stretched, amplified pulse as close as possible to its original duration.

How It Works

The fundamental relationship between the width of a laser pulse and its bandwidth is that a very short pulse exhibits a broad bandwidth, and vice versa. For a Gaussian pulse, this relation is given as

$$d\nu * dt > 0.441 \qquad [1]$$

where $d\nu$ is the bandwidth and dt is the laser pulse width. For example, for a 100 fs duration pulse at $\lambda = 800$ nm, the corresponding bandwidth is more than 9 nm. Therefore, a device that can delay certain frequencies (or wavelengths) relative to others can stretch a short pulse so that it lasts a longer time. Likewise, such a device should also be able to compress a long pulse into a shorter one by reversing the procedure. The phenomenon of delaying or advancing some wavelengths relative to others is called Group Velocity Dispersion (GVD) or, less formally, "chirp."

A pulse is said to have positive GVD, or to be positively chirped, when the shorter (bluer) wavelengths lead the longer (redder) wavelengths. Conversely, if the bluer light is delayed more than the redder light, it has negative GVD or chirp.

For CPA, a combination of dispersive optics are used to form a "pulse stretcher" where low-energy, short-duration pulses can be lengthened by as much as 10^4. Then the energy in these pulses is increased by passing them

through the Ti:sapphire regenerative amplifier. Finally, a set of dispersive optics (similar to those used in the stretcher) are used to form a "pulse compressor" to recompress the pulses to their specified duration. Figure 3-3 illustrates this process.

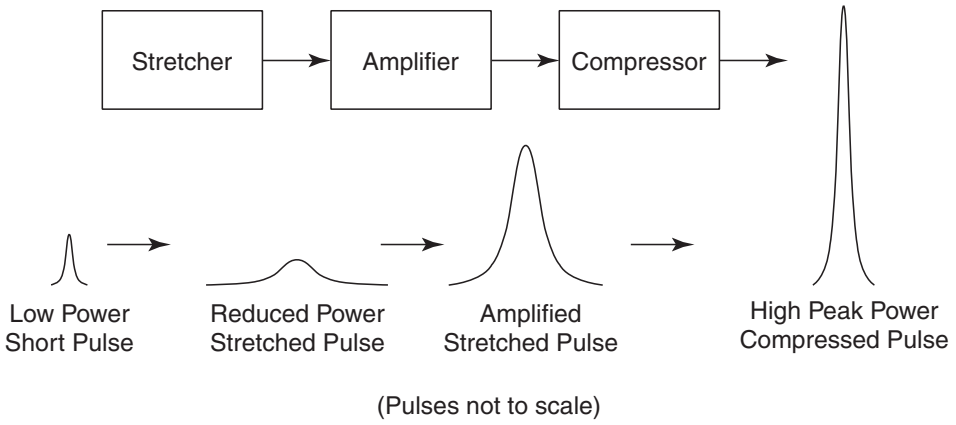


Figure 3-3: The Principle of Chirped Pulse Amplification

Pulse Stretching and Compression

A light pulse incident on a diffraction grating experiences dispersion; that is, its component wavelengths are spatially separated, and so too are its frequency components. The dispersed spectrum can be directed through a combination of optics (usually the same diffraction grating can be used) to send the different frequencies in slightly different directions.

Longer (or redder) wavelengths can be made to travel over a longer path than the shorter (or bluer) wavelengths components of the beam, or vice versa. The result is to lengthen the duration of the pulse, which reduces its peak power (it is the same energy under the curve, only spread out more now).

A prism, which is a simpler optic than a diffraction grating, can also be used for these purposes. However because the pulse passes *through* a prism, negative GVD is introduced by the glass or quartz of the prism body — blue frequencies are delayed relative to the red frequencies each time the pulse passes through the prism. Therefore, gratings are the better choice for CPA because they simplify the process of compensating for dispersion caused by other components in the optical path.

The grating and the routing mirrors can be chosen so that, in the stretcher, the bluer frequency components of the spectrum travel further than the redder components, causing the redder frequency components to exit the stretcher first. In the compressor, the spatially spread beam is flipped so that the redder component have to take the long path, thereby allowing the bluer frequencies to catch up. This recompresses the pulse.

Figure 3-4 shows a simplified pulse stretcher. A short pulse is spectrally spread and then, by making one end of the spread pulse travel farther than the other end, the pulse is temporally broadened.

The same optical components act as a compressor when the leading component of a temporally stretched pulse is forced to take the longer path, thereby allowing the trailing component to catch up. In the pulse stretcher shown below, the bluer components are forced to take the longer path.

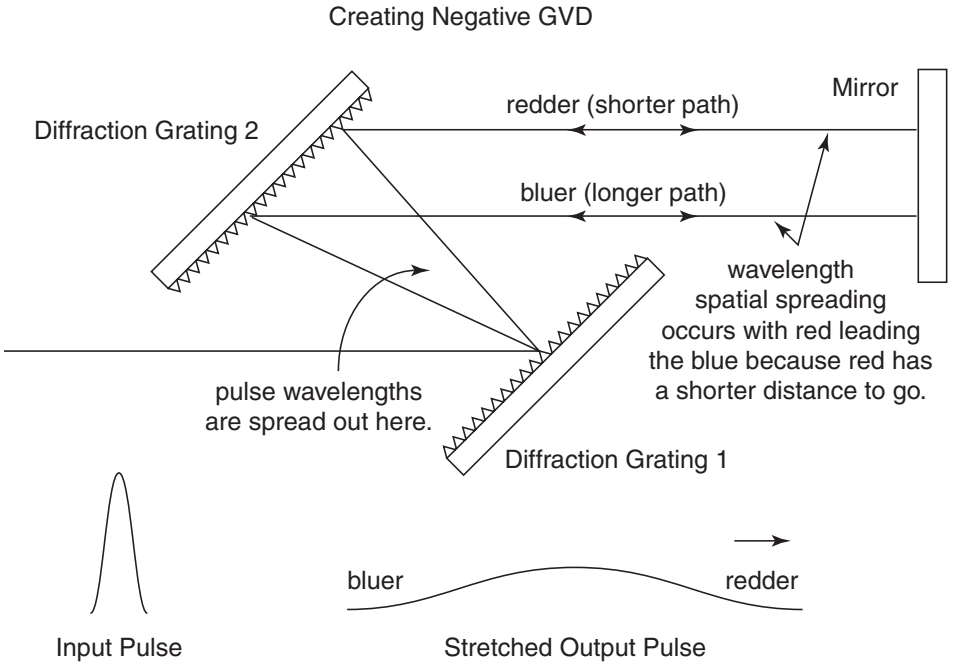


Figure 3-4: Principle of pulse stretching using negative GVD

The Spitfire Pro Pulse Stretcher and Compressor

The design of the *Spitfire Pro* pulse stretcher and compressor makes use of some simplifying modifications. Instead of using two gratings for the stretcher, a simple but elegant retroreflector mirror assembly directs the beam back onto a single grating in the stretcher. This avoids the need to match or to precisely align two stretcher gratings. The beam is also multi-passed to achieve greater spectral spread. This design results in reduced complexity and cost.

The same design principal is used in the compressor, but in reverse. The result is only two gratings are used in the entire system instead of four, simplifying the alignment and maintenance of the system.

If the input to the *Spitfire Pro* is tuned to a different wavelength, the diffraction grating in the stretcher will cause the beam to move, and the grating must be rotated to realign the stretcher. Naturally, the compressor grating must be rotated at very nearly the same angle to ensure optimum pulse compression.

The stretcher and compressor occupy a single chamber and are separated from the regenerative amplifier chamber by an air baffle that minimizes air currents through the stretcher and compressor.

The compressor uses a horizontal retroreflector to flip the red and blue components so that the bluer wavelengths are now forced to take the longer

path. This allows the redder wavelengths to catch up and reduce the pulse duration to close to its original length.

The retroreflector in the compressor is mounted on a track for easy translation in the direction along the beam path. This fine adjustment is used to compensate for small, routine changes in dispersion that take place in the amplifier cavity. Translation control is provided by a motion controller and dc motor.

The design details of the gratings and their optical configuration in a CPA system depend upon, among other factors, the duration of the seed pulses and output pulses. Longer duration pulses have a correspondingly narrower spectrum of wavelengths, and so require a higher density of rulings for the diffraction gratings to achieve an adequate degree of dispersion and stretching.

Each *Spitfire Pro* model has its own stretcher/compressor design, and each model uses gratings with ruling densities that are specific to that model in order to disperse the pulse spectra for that design. The nature of grating diffraction requires each model to have its gratings placed at different angles relative to the beams incident on them. In the *Spitfire Pro F* and *Spitfire Pro PM,* the ruling density of the stretcher grating is the same as that for the compressor grating. The *Spitfire Pro 40F* employs different ruling densities for the stretcher and compressor. In addition, the *Spitfire Pro 40F* and *Spitfire Pro PM* each have different design features for the stretcher and compressor. These features are discussed in further detail below.

The Spitfire Pro 40F Compressor/Stretcher Design

Because the GVD phenomenon described above is not a simple linear effect, extra consideration must be given to a design intended to produce the shortest possible pulses. The frequency components of a pulse traversing an optical system experience dispersion that depends upon the square of the frequency. In addition, dispersion also results from higher order powers of the frequency.

For pulses around 100 fs or longer, any higher-order dispersion resulting from the optical components in the amplifier is small enough so that it is adequately compensated for by the robust design of the *Spitfire Pro* stretcher/compressor. The ruling densities of the stretcher and compressor gratings can be identical because the dispersion introduced in the amplifier has a minimal effect on the duration of the compressed pulse.

However, for pulses of extremely short duration, the higher-order dispersion becomes large enough so that additional compensation is required. The *Spitfire Pro 40F* compensates for this higher-order dispersion by using a “mixed” stretcher and compressor design—the groove densities of the gratings are different. The ruling density of the compressor grating must be increased relative to the stretcher grating to compensate for the additional dispersion introduced by the passage of the ultrashort pulses through the amplifier. This also requires that the stretcher grating and the compressor grating be set at different angles to their incident beams.

The Spitfire Pro PM Compressor/Stretcher Design

Some applications require amplified picosecond pulses rather than femtosecond pulses. While it is possible to produce amplified picosecond pulses from picosecond duration seed pulses, there are advantages to be gained in the stability of the amplified output by producing amplified picosecond pulses from femtosecond seed pulses. In addition, only a mode-locked femtosecond laser (such as the femtosecond *Tsunami*) is required to seed the *Spitfire Pro* amplifier for both pico and femtosecond pulses.

The *Spitfire Pro PM* ("pico-mask") system allows the production of amplified picosecond pulses using a femtosecond seed laser. The stability of this configuration has specific advantages for certain applications, such as pumping an optical parametric amplifier. The *Spitfire Pro PM* converts the pulse spectrum of the femtosecond seed pulses into a spectrum that is equivalent to that produced by a picosecond seed laser.

Recall the relationship between laser pulse width and bandwidth described in "Chirped Pulse Amplification": a very short pulse exhibits a broad spectrum, whereas longer pulses exhibit narrower spectra. Since the pulse stretcher works by spatially separating the spectrum of the seed pulses, the bandwidth of these pulses will be reduced if part of this spectrum is discarded. When the pulse is later compressed, its duration will be longer than if the entire spectrum had been preserved.

The *Spitfire Pro PM* accomplishes this in a straightforward manner. A femtosecond seed pulse enters a stretcher that uses a grating designed for picosecond operation. The spatially spread pulse is then directed onto an aperture that is precisely aligned to mask off part of the spectrum that reflects from the grating, allowing only a portion of the spectrum to pass through. Doing so reduces the bandwidth of the stretched pulse to that produced by a picosecond seed pulse.

If these reduced-bandwidth pulses were allowed to pass through a stretcher designed for femtosecond pulses, optical damage might result. As it is, the bandwidth of these masked pulses is much narrower than the bandwidth of femtosecond pulses—about one nanometer as opposed to ten nanometers.

Picosecond pulses require a greater degree of dispersion in order to produce spatial and, hence, temporal, separation. This is achieved by further increasing the red path length of the beam in the stretcher and increasing the ruling density of the stretcher grating. After amplification, the stretched pulse is directed into the compressor, which is configured just as it would be for a picosecond seed laser (especially in the choice of compressor grating ruling), and amplified picosecond pulses are the result.

A detailed description of the optical layout of the *Spitfire Pro PM* is given in Chapter 7. It is possible to convert one version of a *Spitfire Pro* system to a different version after it has been installed—for example, to convert a *Spitfire Pro F* to a *Spitfire Pro PM*. For for more information about such a conversion, contact your Spectra-Physics representative.

Pulse Selection and Pockels Cells

Once the train of mode-locked seed pulses leave the stretcher, selecting a pulse for retention in the regenerative amplifier is accomplished by exploiting its polarization characteristics. Pockels cells are used to control this polarization and make the pulse selection. A Pockels cell is an electro-optic device that, without a voltage applied to it, has essentially no effect on transmitted light. With a properly applied voltage, however, the crystalline material in a Pockels cell acts as a ¼ waveplate that rotates the polarization of light by 45° each time a pulse passes through it.

If a light beam passes through an active Pockels cell twice (passes through the cell and is then reflected back through it again), the polarization of the beam is rotated by 90°, say from vertical to horizontal.

The *Spitfire Pro* is designed so that horizontally polarized light can be trapped in the regenerative amplifier cavity. The input Pockels cell is paired with a passive ¼ waveplate and the optical path is designed so that the beam makes a double pass through this combination. When the cell is off, the double pass through the passive ¼ waveplate will flip the beam polarization 90°; when the cell is on, the beam experiences a double pass through two ¼ waveplates, leaving its polarization unchanged.

The detailed description of how the input given cell combines with the regenerative amplifier cavity optics to select a pulse for amplification are described in Chapter 7.

The output Pockels cell works in conjunction with the polarizer in the amplifier cavity to release an amplified pulse at a time determined by the *TDG*. Details of how the timing is set for ejecting an amplified pulse are given in the section “The Timing and Delay Generator (TDG)” on page 3-10.

In order for them to work well, the Pockels cells must be first properly aligned with no voltage applied. Likewise the applied voltage must be calibrated to achieve the precise degree of polarization rotation.

One measure of the quality of pulse selection is given by the contrast ratio—the factor by which the amplifier output power exceeds the power in spurious pulses (that are always present to some degree) before or after the main, “true” pulse. The value of the contrast ratio is determined by the quality of the ¼ waveplates, the activation time of the Pockels cells (their intrinsic birefringence) and their drive electronics.

The components selected for the *Spitfire Pro* that affect contrast ratio are of the highest quality available. As a consequence, the limiting factor for contrast ratio is the natural birefringence of Pockels cells. This birefringence results in an optical rise time that is less than a nanosecond. The net effect is high contrast ratios and excellent suppression of spurious pulses.

Note



The 3500 V applied to the *Spitfire Pro* Pockels cells is provided by two high-voltage power supplies inside the *TDG*. The Pockels cells themselves, along with the Pockels cell drivers, are in the *Spitfire Pro* amplifier head.

Regenerative Amplification

A typical laser amplifies the spontaneous emission randomly present in its own gain medium in order to initiate lasing. Regenerative amplifiers, on the other hand, are designed to recirculate and amplify low-energy laser pulses from a separate "seed" laser. Thus, instead of allowing the energy in the amplifier crystal to amplify random spontaneous emission, the seed pulses (having an energy that exceeds the spontaneous emission energy) are selectively amplified. The *Spitfire Pro* can be thought of as a Q-switched, cavity-dumped Ti:sapphire laser that is configured to operate as a regenerative amplifier. Here is how it works:

As explained earlier, Ti:sapphire possesses the broad gain bandwidth that is necessary for the production and amplification of sub-picosecond pulses. In addition, a Ti:sapphire amplifier has a high saturation threshold that makes it possible to extract relatively high energies from a system of moderate size.

A single pass of a very low-energy sub-picosecond pulse through a Ti:sapphire amplifier will increase the pulse energy typically by a factor of about 3 or 4. However, the stimulated emission that provides this gain draws down the population inversion in the gain media only a small amount in a single pass. The Ti:sapphire gain media remains well below the threshold at which stimulated emission will reverse the population inversion (that is, saturate the gain). In short, after only a single pass of a low-energy pulse, there is still a lot of gain left in the amplifier for more passes.

The *Spitfire Pro* cavity is designed to first select and then optically confine an individual pulse from the train of mode-locked seed pulses that have already been lengthened in duration in the stretcher. Reducing the repetition rate from the megahertz mode-locked pulse train to kilohertz rates enables the gain of the amplifier to be concentrated in fewer pulses, thus producing more energy per pulse.

Immediately prior to passing the selected pulse through the Ti:sapphire crystal for amplification, the crystal is excited to population inversion by a high-energy pulse from a separate pump laser. The selected seed pulse is then passed through the crystal 20 or more times until the pulse energy level (the stimulated emission) is high enough to eliminate the population inversion. Having thus saturated the gain, i.e., absorbed all the energy available, the pulse is ejected into the compressor.

Through regenerative amplification, an input pulse of only a few nanojoules of energy may be amplified to roughly a millijoule using a single Ti:sapphire crystal, and multiple passes through the regenerative amplifier can result in an energy amplification greater than 10^6 at the output of the compressor. When the compressor restores the short duration of the pulse, the amplified energy results in correspondingly amplified peak power.

Note



As part of the alignment procedure, the *Spitfire Pro* is sometimes operated as a laser rather than as a regenerative amplifier.

The Timing and Delay Generator (TDG)

The *TDG* provides the timing needed to synchronize the Pockels cells to the passage of the pulses through the amplifier. This allows the Pockels cells to first capture pulses and then, after amplification, direct them into the compressor. This timing includes synchronization to the seed and pump lasers. The *TDG* also provides several additional delays based on the output of the *Spitfire Pro* that allow laboratory instruments to be synchronized to the arrival of pulses at the target.

Immediately after the Ti:sapphire rod is excited by a pulse from the pump laser, the input Pockels cell confines a selected seed pulse in the amplifier where it is directed into the rod for amplification. To do this, the input Pockels cell must be synchronized to the mode-locked pulse train after the next available pump pulse, and remain synchronized after each pump pulse. To achieve this, the input Pockels cell is locked to the RF signal generated by the modelocker in the seed laser. Additionally, the Pockels cell firing phase (the delay) is adjustable to allow the synchronization to be optimized. This ensures that the input Pockels cell fires only after the selected pulse has passed completely through it.

The output Pockels cell ejects the amplified pulse into the compressor. Following the synchronization of the input Pockels cell, there is a delay before the output cell is activated to ensure the captured pulse is released at optimum amplification. This delay is adjustable from 0 to 1000 ns, which allows the pulse to complete the amplifier cavity path an integral number of times.

The *TDG* is first triggered by a TTL positive edge pulse provided by the pump laser. It then produces separate triggers with adjustable delays for both Pockels cells. The PC1 control on the main screen of the *Spitfire* GUI (corresponding to CH1 on the back panel of the *TDG*) connects to the input Pockels cell; PC2 on the main screen (corresponding to CH2) connects to the output Pockels cell. These controls are used to adjust the corresponding delays, in nanoseconds.

The SYNC control is adjusted to synchronize target or monitoring devices to the *Spitfire Pro* output pulse. As a simplified example, SYNC can be used to provide horizontal triggering for an oscilloscope.

Note



The repetition rate of Pockels cell switching (and hence the repetition frequency of the *Spitfire Pro* output) is dependent on the repetition rate of the input trigger from the pump laser.

The *TDG* contains high-voltage power supplies that power the Pockels cells. The *TDG* also contains the controls and signals for the Bandwidth Detector (BWD) that protects the amplifier optics from damage caused by improper seed pulse input.

The design and the control of the BWD is explained in Chapter 4, “Controls, Indicators and Connections,” along with instructions for operating the *TDG*.

Specifications

The *Spitfire Pro* amplifier systems are available in a number of configurations. The tables below show the configurations covered in this manual.

Note



The specifications in Table 3-1 only apply when the *Spitfire Pro* is pumped and seeded using the recommended Spectra-Physics lasers. The models listed in Table 3-1 are pumped by the *Empower-15* laser.

Table 3-1: *Spitfire Pro* Specifications[1] by Model

Amplifier Model[2]	Output Energy[3]		Pulse Width[4]	Pre-Pulse Contrast Ratio[5]		Wavelength[6] (nm)
	1 kHz	5 kHz		1 kHz	5 kHz	
F	> 1 mJ	> 0.3 mJ	< 130 fs	1000:1	500:1	750–900
PM	> 1 mJ	> 0.3 mJ	1–2 ps	1000:1	500:1	750–900
40F	> 0.7 mJ	> 0.2 mJ	< 40 fs	1000:1	500:1	780–820

[1] *Due to our continuous product improvement program, specifications may change without notice. Specifications listed on the purchase order supersede all other published specifications.*
[2] *Designators "1K" and "5K" refer to repetition rates of 1 kHz and 5 kHz, respectively. If optimum performance is required at more than one repetition rate, an additional optic set is required. Any system can be operated with the same energy per pulse at reduced repetition rates through the divide-down electronics on the TDG.*
[3] *Output energy per pulse: applies between 780 and 800 nm. For higher energy output systems, please contact Spectra-Physics.*
[4] *Pulse width applies at the peak wavelength and requires the seed laser performance specified for the Amplifier model. A Gaussian pulse shape (0.7 deconvolution factor) is used to determine the pulse width (FWHM) from an autocorrelation signal as measured with a Spectra-Physics Model PulseScout.*
[5] *Contrast ratio is defined as the ratio between the peak intensity of the output pulse to the peak intensity of any pulse that occurs more than 1 ns before the output pulse. The contrast ratio for any pulse more than 1 ns after the output pulse ("post-pulse contrast ratio") is > 100:1. For higher performance, please contact Spectra-Physics.*
[6] *Wavelength: the system is tunable from 750 to 840 nm without an optics change. To cover the 840–900 nm region, separate optics are required for both 1 K and 5 K systems. For other wavelengths and for second and third harmonic generation, please contact Spectra-Physics.*

Table 3-2: *Spitfire Pro* Specifications Common to All Models

Beam Diameter[1]		Beam Divergence[2]	Transform Limit[3]	Energy Stability[4]	Output Polarization
Base Models	HPR Models				
7 mm	12 mm	<1.5	<1.5	<1%	horizontal

[1] *Nominal beam diameter at $1/e^2$ points.*
[2] *TEM_{00}. Beam divergence as a multiple of the diffraction limit.*
[3] *Assuming seed pulses are transform-limited Gaussian temporal pulses.*
[4] *RMS value at peak wavelength between 780 and 800 nm.*

Outline Drawings

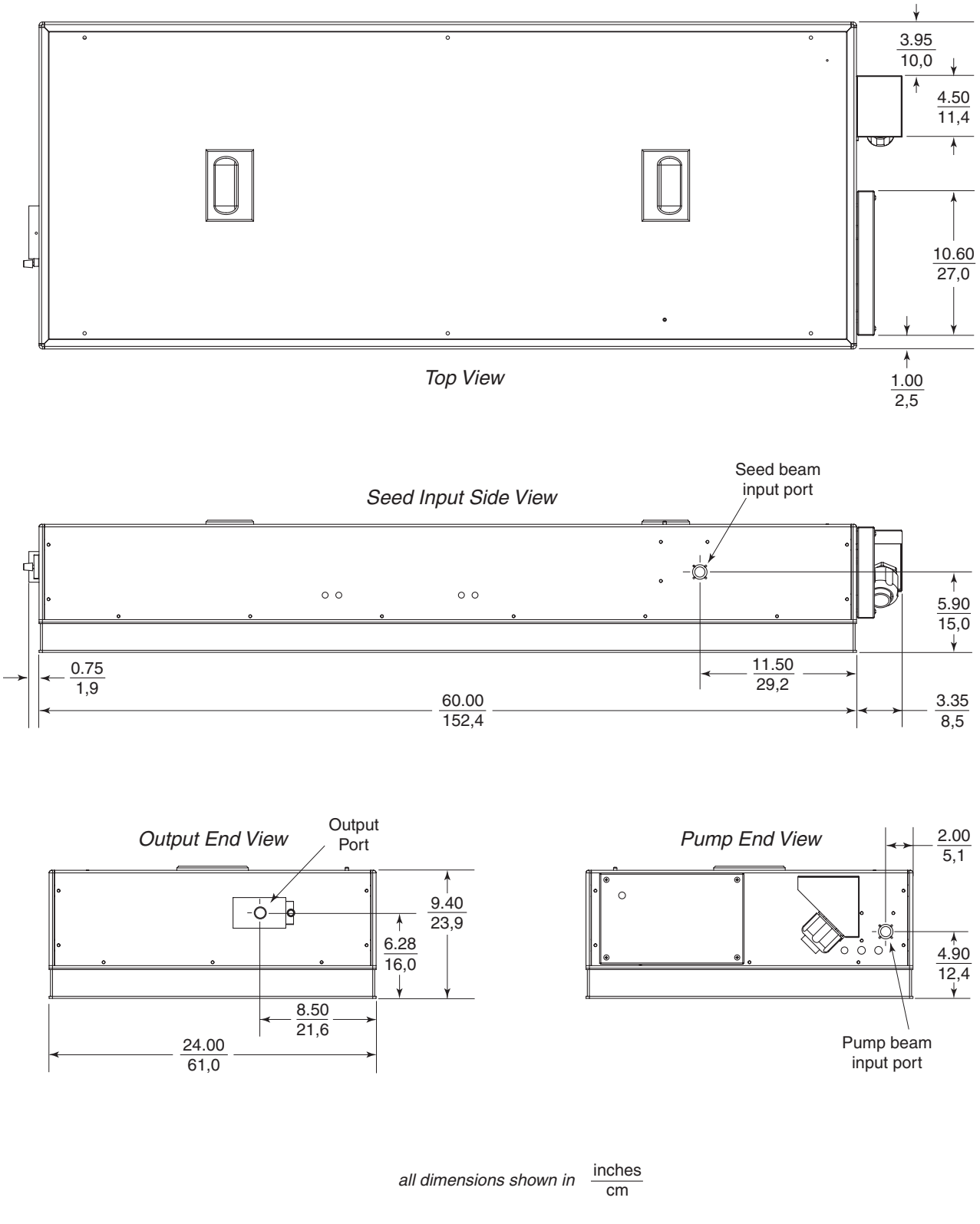


Figure 3-5: *Spitfire Pro* Outline Drawing

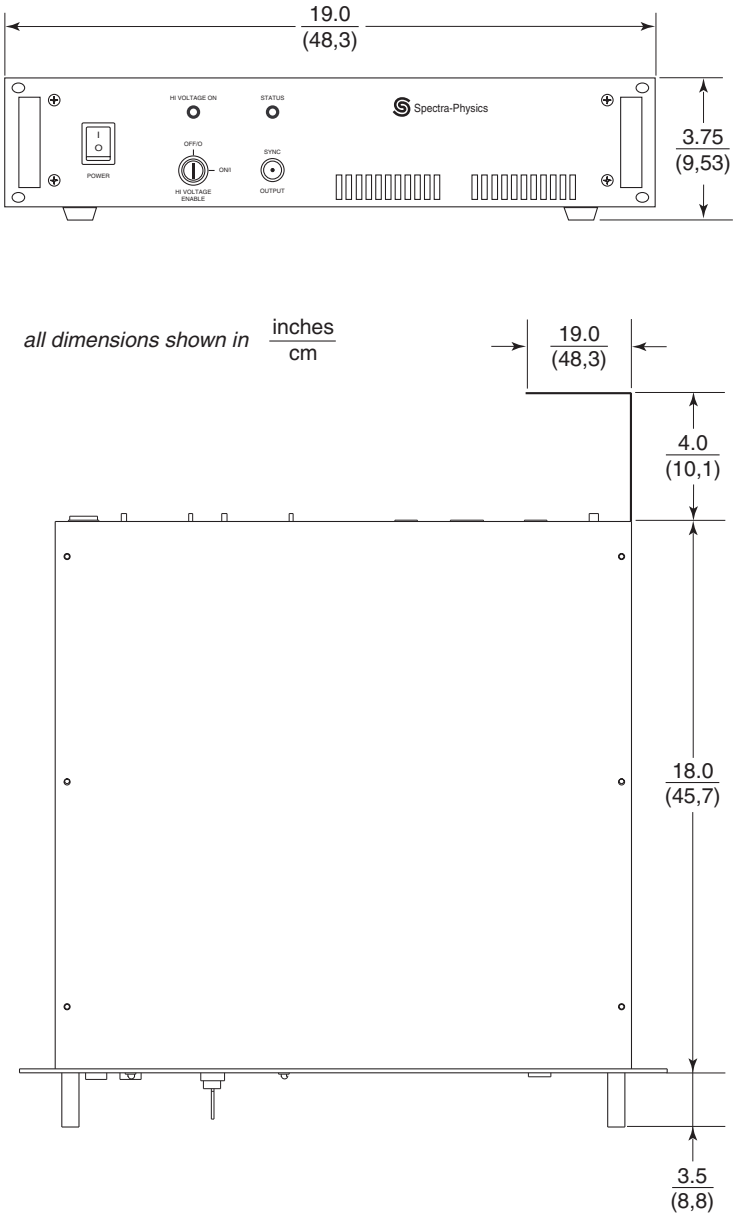


Figure 3-6: *TDG* Outline Drawing

Chapter 4 Controls, Indicators and Connections

This chapter describes the controls, indicators and connections on the external panels of the *Spitfire Pro* system. It describes the external panels of the amplifier head, the *Timing and Delay Generator* (*TDG*) and auxiliary connections. Understanding the function of these mechanisms is necessary in order to operate the system properly. System control is via the custom graphical interface as described in Chapter 6.

Spitfire Pro Head External Controls

Pump Input End Panel

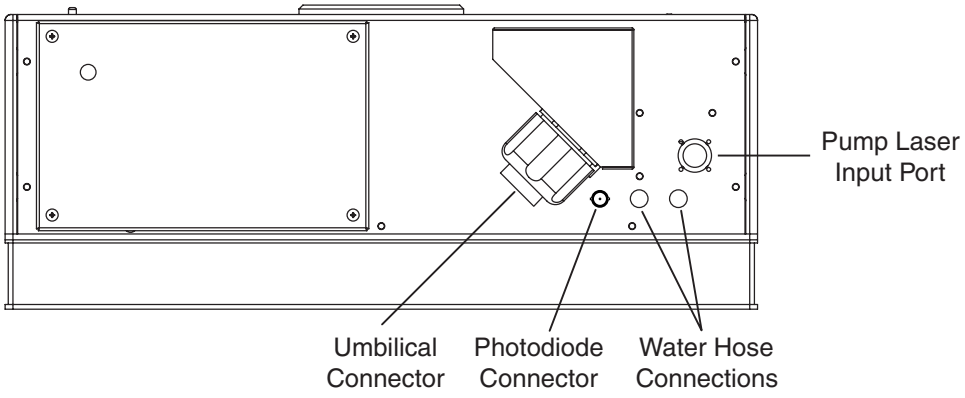


Figure 4-1: *Spitfire Pro* Pump Input Panel

Pump laser input port —provides entrance for the pump beam (e.g., from a Spectra-Physics *Empower-15* laser).

Umbilical connector—provides attachment for umbilical containing the laser communication cable, Pockels cell high voltage cable, and CH1 and CH2 Pockels cell trigger cables.

Photodiode connector—provides connection to the photodiode behind on the amplifier cavity mirrors. This detector measures a fraction of the amplified light that passes through the mirror for use in alignment and troubleshooting.

Cooling water connections—provide cooling water for the amplifier rod. Water is shared in parallel with the *Empower* pump laser. Either connector can be used as the IN or OUT connection for the water flow.

Seed Input Side Panel

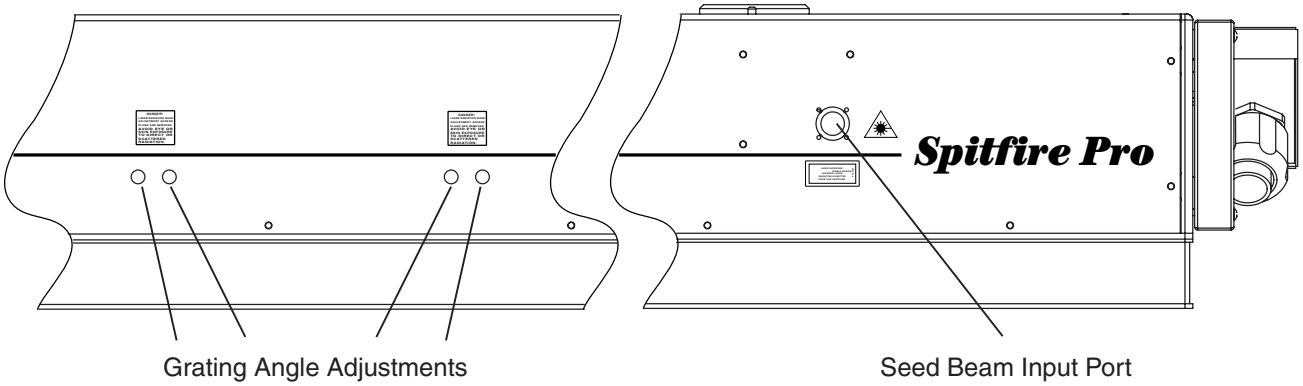


Figure 4-2: *Spitfire Pro* Seed Input Side Panel

Seed Beam Input Port—provides entrance for the seed beam (from a *Tsunami* or *Mai Tai* mode-locked laser).

Grating Angle Adjustment (not labeled)—provides adjustment for the angle between the stretcher diffraction grating and the incident seed beam, and the angle between the compressor diffraction grating and the amplified beam. Two pairs of access holes for adjustment are provided; only one pair of adjustments is used in each model of the *Spitfire Pro*, which have different locations for the gratings.

These adjustments are to be used only by service technicians trained in their use by Spectra-Physics.

Output End Panel

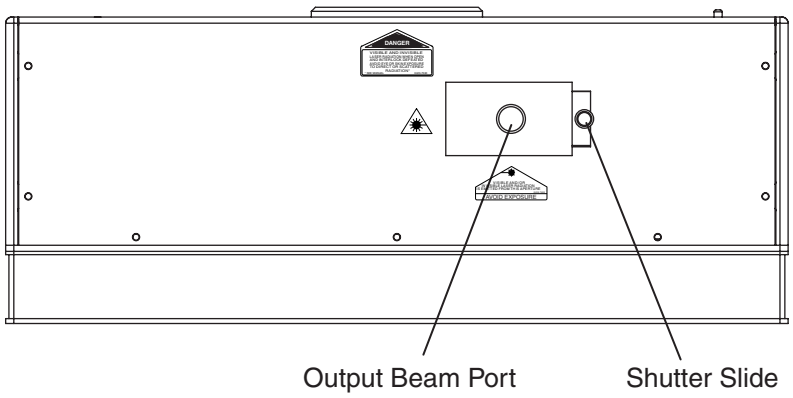


Figure 4-3: *Spitfire Pro* Output End Panel

Output Beam Port—provides an exit for the amplified output laser pulse.

Shutter Slide—controls the shutter for the output beam. When closed, it blocks the beam.

The Timing Delay Generator (TDG)

The *TDG* controls the selection of pulses from the seed laser and the repetition rate of the pulsed output of the *Spitfire Pro*. The *TDG* counts and then selects mode-locked seed pump pulses at either the 1 kHz or 5 kHz amplifier rate, or a lower rate selected by the user.

The *TDG* also synchronizes the seed pulses with pulses from the pump laser—it captures the next seed pulse while the laser rod is still excited by the pump pulse. It does this by providing an adjustable delay (in nanoseconds) to which the input Pockels cell can be set to capture the pulse.

The second adjustable delay controls the output Pockels cell to eject the pulse into the compressor after it has been amplified. The *TDG* allows the output repetition rate to be reduced from its pre-set value by dividing the input synchronization signal from the pump laser. Preset integer divider values are provided.

Additional adjustable delays provide triggers for laboratory equipment such as the horizontal sweep of a high-speed oscilloscope. The connectors for these additional delay lines are on the *TDG* back panel.

The *TDG* also contains the high-voltage power supplies that drive the Pockels cells. The drivers themselves are located in the *Spitfire Pro* above the regenerative amplifier cavity.

The *TDG* is controlled by the *Spitfire Pro* GUI control software.

Front Panel

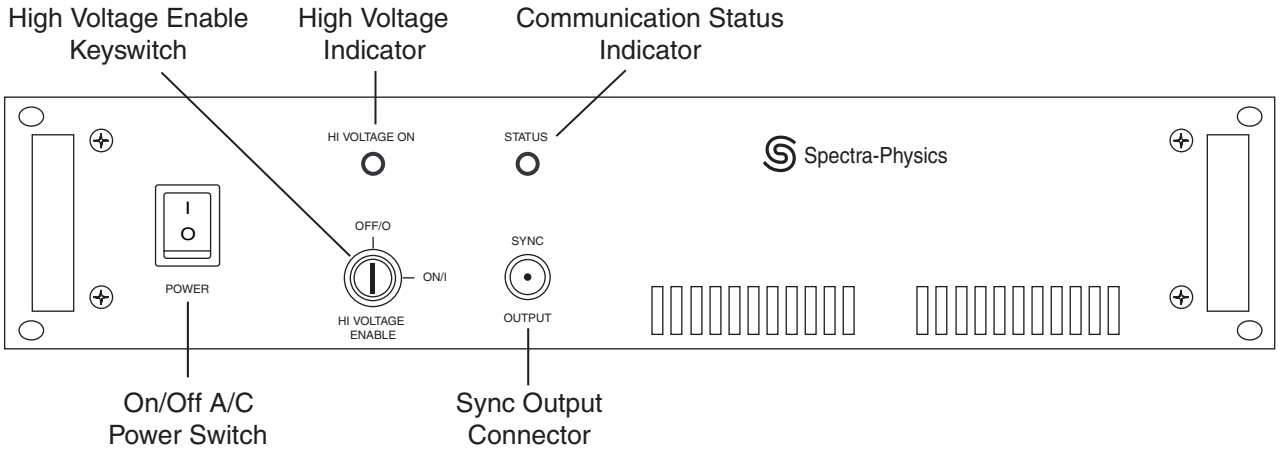


Figure 4-4: *TDG* Front Panel

POWER switch—turns ac power on (**I**) and off (**O**).

HI VOLTAGE ON indicator—illuminates when the *TDG* is supplying high voltage to the amplifier head.

HIGH VOLTAGE ENABLE keyswitch—enables (**I**) or disables (**O**) the high voltage output to power the Pockels cells in the amplifier head.

STATUS indicator (green)—illuminates when the communication link with the computer control is functioning properly.

SYNC OUTPUT connector (BNC)—provides attachment for a timing signal synchronized to the pulsed output of the *Spitfire Pro* for controlling laboratory instruments (e.g., an oscilloscope). The timing signal has an adjustable delay controlled through the *Spitfire* Main menu.

The Bandwidth Detector (BWD)

The BWD protects the regenerative amplifier optics from damage if the stretcher cannot adequately reduce the peak power of the seed pulses before they are amplified. Inadequate stretching may occur, for example, if a portion of the beam in the stretcher is blocked.

The BWD relies on the signals from two fast photodetectors that capture one of the reflections from the stretcher grating. If the signal from either photodetector falls below a threshold (factory set for each version of the *Spitfire Pro)*, the BWD is activated, which disables the trigger signals that fire the Pockels cells. No pulses are selected for amplification, and the optical components are, thus, protected. When the seed laser is stable and properly mode-locked, the BWD permits the *TDG* to function normally.

Note

Since the angle of the stretcher grating varies for different models of the *Spitfire Pro*, the location of the BWD photodiodes also vary.

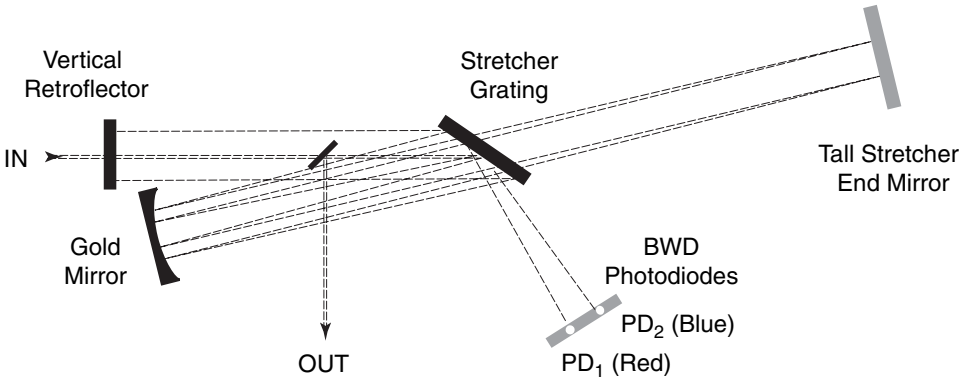


Figure 4-5: Optical Layout of the BWD in the *Spitfire Pro F*. For clarity, some amplifier components are not shown.

Warning!

The BWD must function properly in order for it to protect the *Spitfire Pro* amplifier optical components from permanent damage. Such damage is expensive to repair. If making internal adjustments while the amplifier is operating, make sure the seed beam is not partially blocked in the stretcher.

Back Panel

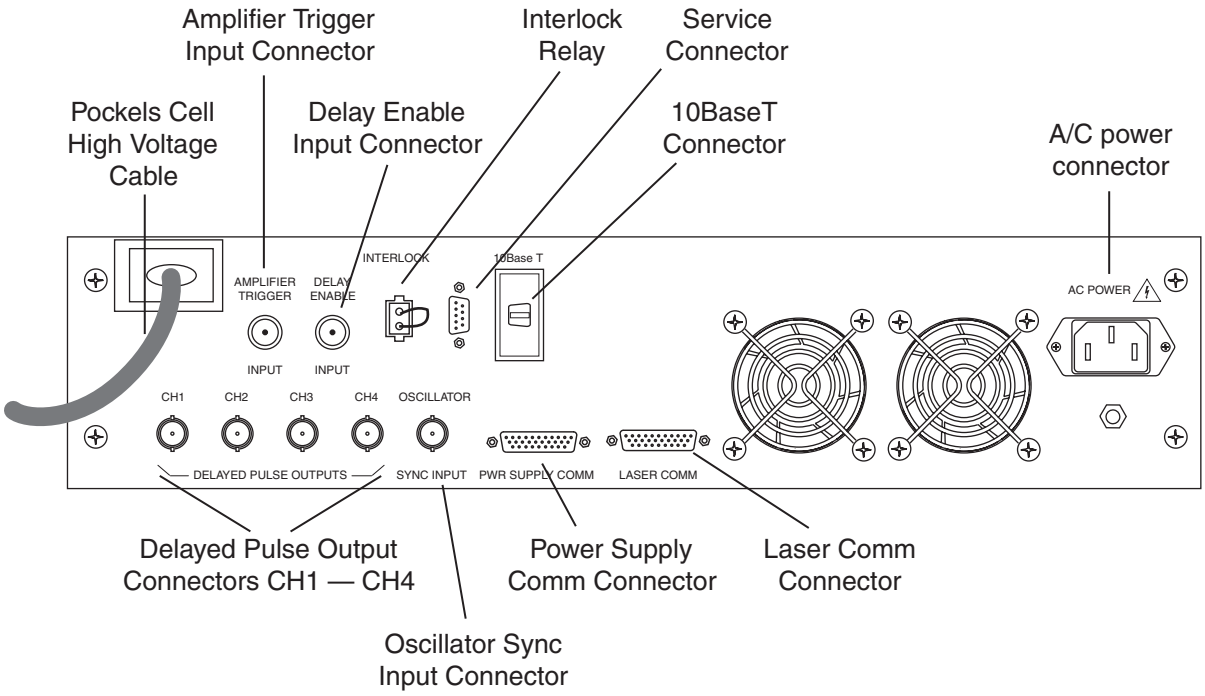


Figure 4-6: *TDG* Back Panel

Pockels Cell High-Voltage Cable (unlabeled) — provides high-voltage power to the Pockels cells in the *Spitfire Pro* amplifier head.

AMPLIFIER TRIGGER INPUT connector (BNC) — connects to the SYNC OUTPUT connector of the *Empower* power supply to synchronize pump input.

DELAY ENABLE INPUT connector (BNC) — reserved for Spectra-Physics use.

INTERLOCK relay (2-pin) — provides a means to quickly shut off amplifier output if a user-installed safety switch is opened. To use this interlock, remove the factory-supplied jumper plug from the panel and replace it with a similar connector that is wired to an external safety switch. The switch can be attached to an access door or to other auxiliary safety equipment. Wire the switch as "normally closed" so that when the door or safety device is opened, the switch opens and the high-voltage to the *Spitfire Pro* Pockels Cells is immediately turned off, disabling amplified output.

Note that this interlock will disable the *Spitfire Pro* amplified output only. It will not disable the pump or seed lasers. Those lasers are equipped with interlocks that may be similarly wired as switches. Do not tie the various interlocks together electrically; they are separate circuits. Use separate switches, or use separate poles on the same switch.

SERIAL connector (9-pin D-sub) — reserved for Spectra-Physics use.

10BaseT connector — provides attachment for an Ethernet connection to the computer running the *Spitfire* GUI control software. Settings for this port are provided in Chapter 5, "Installation."

AC POWER connector (110/220 Vac) — is the ac power input for the *TDG*. The *TDG* is autoranging for 110–220 Vac service. The connector includes EMI protection and either a 6 A (110 Vac) or a 3 A (220 Vac) fuse.

DELAYED PULSE OUTPUTS (BNC)

CH1, CH2 — correspond to the PC1 and PC2 controls on the Main menu. They are used to synchronize timing of the capture and release of the pulses into and out of the regenerative amplifier.

CH3, CH4 — correspond to the CH3 and CH4 controls on the Timing and High-Voltage menu. These are optional timing signals that are provided to synchronize laboratory equipment.

OSCILLATOR SYNC INPUT connector (BNC)—connects via a high-speed BNC cable to the modelock synchronization output on the seed laser. If a *Mai Tai* is used, connect this *TDG* input to the 40 MHZ output connector (NOT the 80 MHZ output connector); for a *Tsunami* seed laser, connect it to the MONITOR output on the *Tsunami* controller (NOT the SYNC output). Refer to the appropriate user's manual for your seed laser.

Jitter is specified at <250 ps. Input impedance is 50 Ω, internally switchable.

PWR SUPPLY COMM connector (26-pin D-sub) — connects to the RS-232 connector on the power supply of the *Empower* pump laser. It provides control and monitoring of the pump beam into the *Spitfire Pro*.

LASER COMM connector (26-pin D-sub) — provides attachment for the D-sub connector on the umbilical communication cable. It provides control and monitoring of the *Spitfire Pro* amplifier.

Chapter 5 Preparing for Installation

Caution!

Call your Spectra-Physics service representative to arrange an installation appointment, which is part of your purchase agreement. *Allow only an authorized Spectra-Physics representative to install your Spitfire Pro system.* You will be charged for repair of any damage incurred if you attempt to install the *Spitfire Pro* yourself, and such action may void your warranty.

System Components

Because a typical installation requires both a pump laser and a seed laser in addition to the *Spitfire Pro*, some planning is required before beginning installation. Typical system components include:

- *Empower-15* a multi-kilohertz, intracavity doubled, diode-pumped Nd:YLF pump laser

and a

- *Mai Tai* femtosecond Ti:sapphire, modelocked seed laser (this system includes its own internal diode-pumped, CW pump laser)

or a

- *Tsunami* femtosecond or picosecond Ti:sapphire, mode-locked seed laser

together with a

- *Millennia* diode-pumped, CW laser for pumping the *Tsunami.*

Note

Although not recommended, it is possible to use other seed or pump lasers as components in a *Spitfire Pro* system. However, the use of a seed laser other than the *Mai Tai* or *Tsunami* will likely require a pre-collimation procedure to avoid the introduction of spatial chirp in the stretcher.

Also included:

- a Windows®-based laptop computer with the *Spitfire* GUI control software installed
- the *Timing and Delay Generator* (*TDG)*, which can be mounted in a standard 19 in. rack.

Windows is a registered trademark of the Microsoft Corporation

Pump Laser

The *Spitfire Pro* is designed for optimum performance when pumped by a Spectra-Physics *Empower* laser, a frequency-doubled Nd:YLF laser. Care has been taken to match the *Spitfire Pro* optics to this pump laser, especially with regard to wavelength, beam diameter, divergence and the ability to focus the input beam into the *Spitfire Pro* Ti:sapphire rod.

We recommended the *Spitfire Pro* be pumped only with a Spectra-Physics *Empower* laser. If this is not the case, your *Spitfire Pro* warranty may be voided unless prior written approval is obtained from Spectra-Physics.

The pump laser must meet the following specifications:

Table 5-1: Pump Laser Specifications

	1 kHz	5 kHz
Energy per pulse (mJ)	10	3.0
Average Power (W)	10	15
Wavelength (nm)	527	527
Beam Diameter (nominal)	3 mm	3 mm
Energy Stability (±p/p)	<2%	<3%
Beam Profile	Multi-mode, uniform intensity	
Polarization	Linear horizontal	

Versions of *Spitfire Pro* amplifiers other than those listed in this manual may be pumped by other lasers. In addition, previous *Spitfire* systems may likewise use earlier versions of pump lasers. Contact your Spectra-Physics representative for more information.

Modelocked Seed Laser

The *Spitfire Pro* is designed to work with a *Tsunami* or *Mai Tai* modelocked Ti:sapphire seed laser. These are exceptionally stable systems. Spectra-Physics is not responsible for problems caused when a laser other than one of these is used to seed the *Spitfire Pro* laser.

The seed laser must meet the following specifications:

Table 5-2: Seed Laser Specifications

Wavelength	750–950 nm
Power	>400 mW
Beam Diameter at 1/e^2 points	<2 mm
Stability	<1% rms
Pulse Length	<85 fs (Spitfire Pro F) <85 fs (Spitfire PM) <30 fs (Spitfire Pro 40F)
Polarization	Linear vertical
Beam Divergence, full angle	<0.6 mrad

Preparation

Location and Layout

Each user will probably have unique layout requirements based on the application and the requirements and layout of the experiment, so when choosing a layout, please consider the following:

- The *Spitfire Pro* head covers an area of table space equal to 5 x 2 ft (1.9 x 0.75 m), plus additional space for the *TDG* if it is not rack-mounted or otherwise placed elsewhere off the table. Allow sufficient space around the assembly for water hoses, high-voltage connections, etc.
- Position the *TDG*, the control computer and the power supplies for the pump and seed lasers so that the connecting cables easily reach their connectors without strain. Generally, the components should be within about 2 meters of each another.
- Select a location where the electrical utilities for all the laser systems are readily available. Spectra-Physics strongly recommends that the laser system be located in a laboratory environment, i.e., a room that is free from dust and drafts and does not exhibit any large temperature fluctuations. Room temperature should be maintained to within ±2°C during operation.
- For stability, the entire system should be placed on a single, standard optical table.
- Because occasional adjustments might be required to optimize performance, position the *Spitfire Pro* to allow easy access to its internal controls.
- Place the seed laser as close as possible to the *Spitfire Pro* to avoid beam instability problems (such as those caused by unstable routing mirrors or by too many mirrors). Only use stable routing mirrors.
- Enclose all of the laser beams with beam tubes.
- Both the pump laser and mode-locked seed laser must operate within the specifications listed earlier.

The *Spitfire Pro* is shipped pre-assembled, but some optics may have been removed and carefully wrapped for protection during shipment. Leave them wrapped at this time. The Spectra-Physics representative assigned to perform the initial installation will unwrap and install these optics.

Required Utilities

The *Spitfire Pro* requires access to either 110/120 Vac 15 A, or 220/230 Vac 10 A single-phase power. The seed and pump laser systems have electrical and cooling requirements as well. Before beginning installation, refer to the user manuals for those units.

Note



Make certain proper utility service is available at the site before the Spectra-Physics field technician arrives for the initial installation.

Recommended Diagnostic Equipment

The following equipment is recommended for day-to-day operation of the *Spitfire Pro*:

- a power meter capable of measuring between 10 mW and 20 W average power from 500 nm to 900 nm (e.g., a Spectra-Physics 407A)
- a fast CRT analog oscilloscope capable of 300 MHz or better (e.g., a Tektronix 2467, 7104 or 2465)
- a fast photodiode with a 2 ns rise time or better (e.g., an Electro-Optics Technology Model ET 2000)
- IR viewer and IR card
- an autocorrelator (e.g., a Spectra-Physics PulseScout)

Tools Required:

The following tools may be needed during installation, maintenance and/or troubleshooting:

- a small, low-divergence HeNe laser (for alignment)
- three gimbal mounts with 4−6 in. adjustable height
- alignment pins
- 10 in. (25 cm) scale
- three silver mirrors for the above mounts
- #1 Phillips screwdriver
- white business card
- reagent-grade methanol
- lens tissue
- standard, U.S. hex ball-driver set
- gel linear polarizing film

Interconnect Diagram

Figure 5-1 is a schematic representation of the main signal and control connections between components of the *Spitfire Pro* system. For clarity, the more obvious connections are not shown (ac power for the *TDG,* for example).

Also not shown are the power, water, and control connections for the pump laser and seed laser. Refer to the *Mai Tai* or *Tsunami* (seed laser) and the *Empower-15* (pump laser) user's manuals for this information.

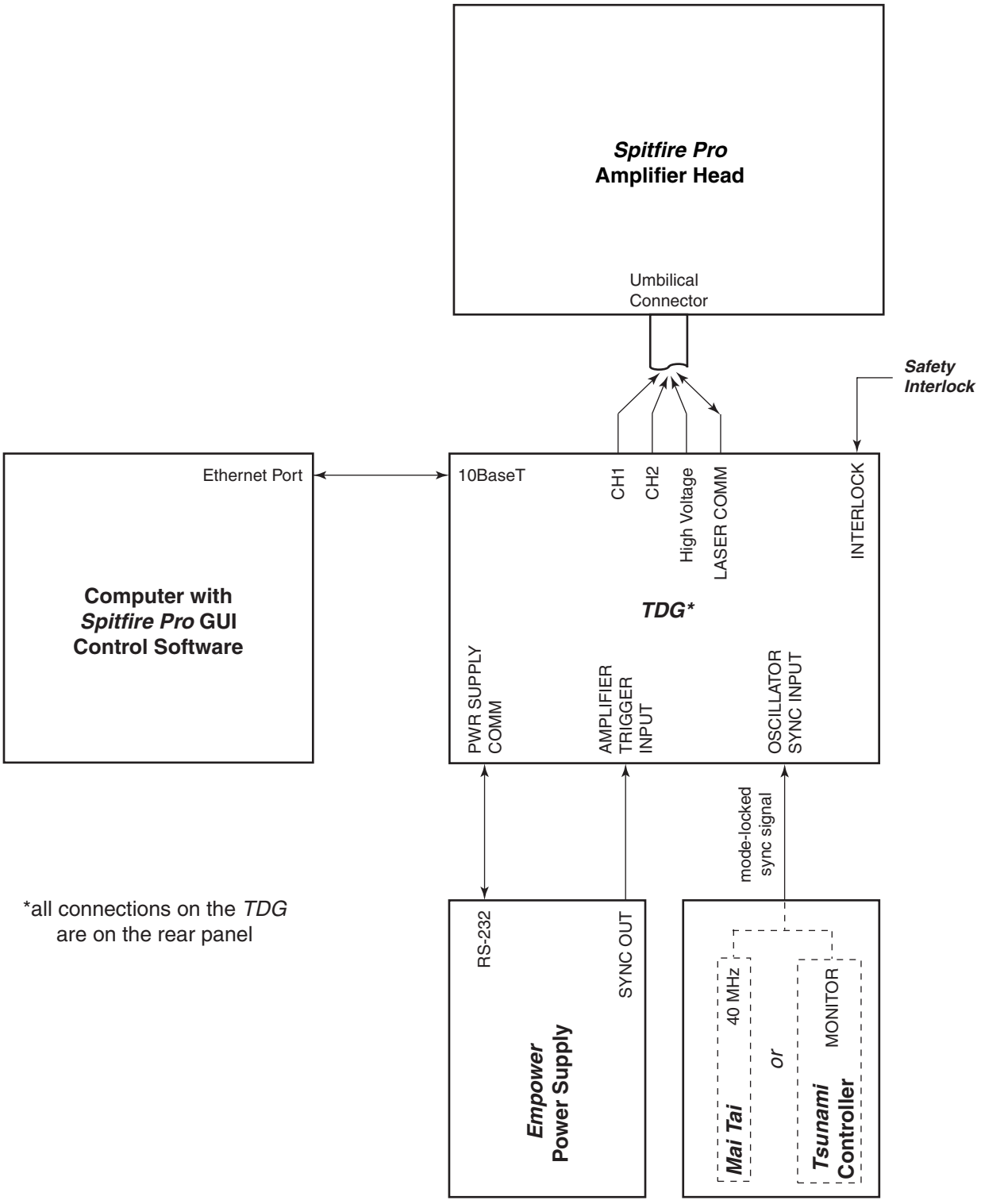


Figure 5-1: *Spitfire Pro* Interconnect Diagram

Chiller

The *Spitfire Pro* Ti:sapphire amplifier rod must be cooled to avoid damage. The cooling water is shared in parallel by the amplifier and the *Empower* pump laser as shown in Figure 5-2.

Note that the chiller for the seed laser operates independently of the *Spitfire Pro* system.

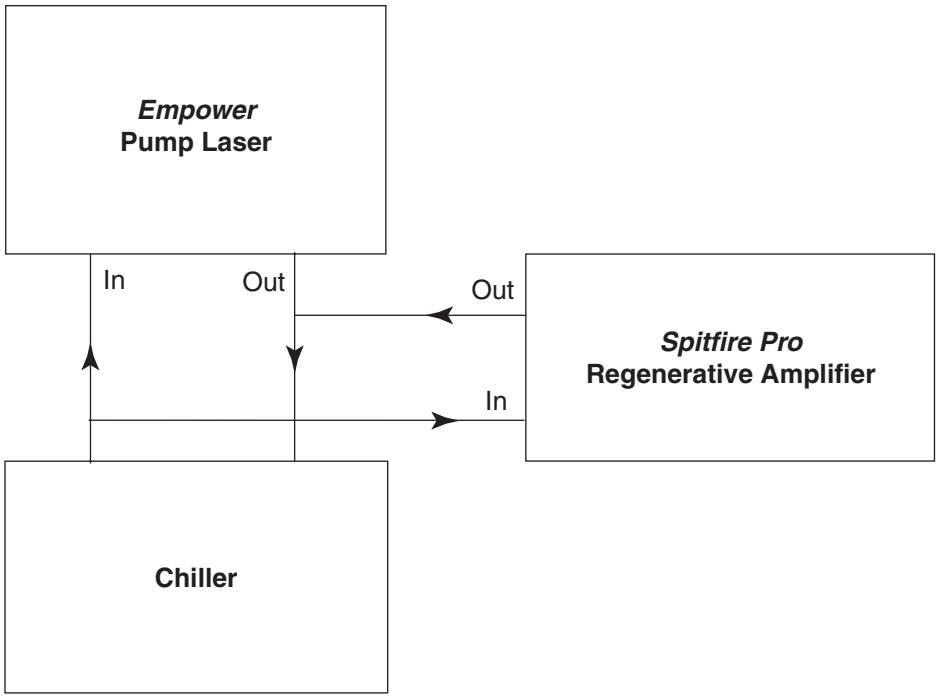


Figure 5-2: Parallel Connections for Chiller Water for the *Spitfire Pro*

Setting the Computer IP Address

In order to communicate with and control the *Spitfire Pro*, the Ethernet port on the computer running the *Spitfire* GUI must be set to a "fixed IP address." If the laptop shipped with the system is used, this has already been done. If a different computer is used, perform the following to insure communication with the *Spitfire Pro* system. (Note: a fixed IP address is *not* a Windows (XP, 2000, ME) operating system default setting.) To set this port:

1. From the Windows Start menu, choose "Settings > Control Panel > Network and Dial-up Connections."
2. If necessary, disable all Ethernet ports except the one that is going to be used (right click on each port listed, then choose "Disable"). Stand-alone computers typically have only one Ethernet port.
3. Verify the Ethernet port to be used is enabled (right click on that port and choose "Enable").
4. Right click again on the port to be used, and choose "Properties."
    a. Highlight "Internet Properties"
    b. Click on "Properties."
    c. Click on the radio button marked "Use the following IP Address."
    d. Set the IP Address to 10.19.70.1
    e. Set the Subnet mask to 255.255.255.0
5. Click "OK" several times to close all the windows.

The Ethernet port is now set to communicate with the *Spitfire Pro* system.

Chapter 6 Operation



Laser radiation is present. Safety glasses of OD 4 or greater at all lasing wavelengths must be worn at all times when operating this laser system.



The following procedures assume you are familiar with the safety information in Chapter 2, as well as with the controls and indicators of the *Spitfire Pro* software menus described at the end of the this chapter. If you have not already done so, read this information thoroughly before attempting to operate the *Spitfire Pro*.

It is recommended that the following equipment be kept on hand:

- a power meter capable of measuring between 10 mW and 20 W average power from 527 nm to 900 nm (e.g., a Spectra-Physics 407A)
- a fast photodiode with a 2 ns rise time or better
- a fast CRT analog oscilloscope capable of 300 MHz or better
- IR viewer and IR card
- an autocorrelator (e.g., a Spectra-Physics PulseScout)

Start-up Procedure

1. Turn on the seed laser system (*Mai Tai* or *Tsunami*), including its chiller, as described in the seed laser user's manual. This may include turning on the pump laser (*Millennia*) for the seed laser.
2. If necessary, turn on the chiller for the *Empower.*
3. If necessary, turn on the ac power to the *Empower* power supply and the *TDG*. Note that, if the *Empower* power supply was off, it will need about 10 minutes to heat the LBO doubling crystal to its proper operating temperature.
4. If necessary, start the *Spitfire Pro* control computer and log on. Start the *Spitfire Pro* control software.
5. Verify both BWD indicators (PD1, PD2) on the Main menu are green. If not, permit more time for the seed laser to stabilize.
6. On the front panel of the *TDG*, turn the HI VOLTAGE keyswitch to the ON position.
7. Turn the keyswitch on the front of the *Empower* power supply to the "on" position.

8. Verify the MAIN, EMP, AMP, and TDG communications indicators on the Main menu are green.
9. Verify the STOP indicator is green.
10. Open the Empower Setup menu window (View > Empower Setup).
11. Set the SET CURRENT control to the correct value (either the current value in use when the system was last shut down, or the value shown on the Ship Report that was sent with the system).
12. Close the Empower Setup menu and navigate to the Main menu.
13. Click on and hold in the green RUN button on the Main menu until the laser starts (about 5 seconds). See the description of the Main menu later in this chapter for a detailed description of the start sequence performed by the system at this point. Allow a 15 minute warm-up period.
14. Enable the two Pockels cells by clicking on the buttons below PC1 and PC2 on the Main menu.
15. Verify the *Spitfire Pro* is producing amplified pulses at the correct timing.
16. Use the motor controls on the Main menu to optimize the compressor length to produce the shortest duration pulses. Refer to "Optimizing Pulse Compression" in this chapter for instructions.
17. Adjust the *Spitfire Pro* repetition rate using the DIVISOR control on the Timing and High Voltage menu (if so desired).

Shut-down Procedure

1. Before shutting down, record the *Spitfire Pro* output power into a system log, along with the current level of the *Empower* pump laser and the timing parameters of the Pockels cells.
2. Press the red STOP button on the Main display.
3. Exit the *Spitfire Pro* software.
4. Turn off the *TDG* and the *Empower* keyswitch.
5. If the system is not going to be used for an extended period of time, turn off ac power to the *TDG* and the *Empower*, otherwise leave power applied (this keeps the crystals warm).
6. Turn off the chiller if the *TDG* and the *Empower* were turned off, otherwise, leave it on.
7. Power down the seed laser (*Mai Tai* or *Tsunami)* as described in their user's manual. The chiller for the seed laser should always remain on.

Basic Performance Optimization

Optimizing Pulse Compression

Temperature changes or similar variations in the *Spitfire Pro* environment may require adjustment of the compressor to optimize the pulsed output.

Use the MOTOR SPEED control on the Main menu to set the compressor length for optimum compression. This adjustment is critical for femtosecond operation: the length must be within about 0.1 mm of the optimum. The best way to set the compressor length is to monitor the pulse width using an autocorrelator while using the MOTOR SPEED control to adjust the horizontal retroreflector.

The compressed pulse should look like that shown in Figure 6-1:

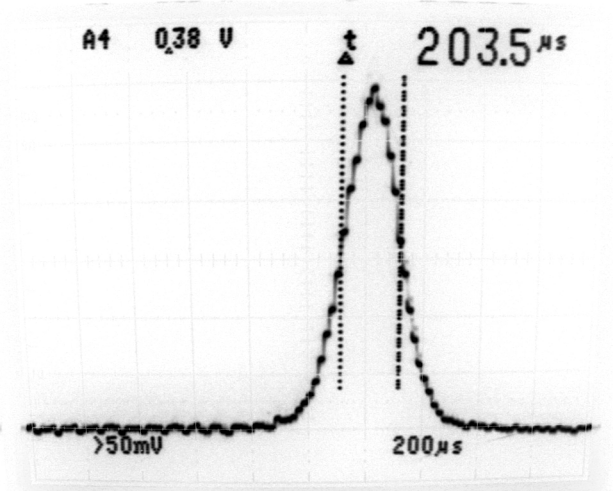


Figure 6-1: Autocorrelation of a Well Compressed Pulse

It is possible to optimize the pulse length by observing the amplifier output on a white business card. When the compressor length is correct, the beam on the card will appear blue in the center due to high peak power frequency doubling in the treated paper.

The following parameters do not need to be optimized on a daily basis. Nevertheless, they are fundamental to proper operation of the system:

- stability of the seed pulses
- seed beam alignment into the regenerative amplifier
- beam uniformity
- build-up reduction time (optimizing the regenerative amplifier)

Stability of the Seed Pulses

The mode-locked output of the seed laser must be optimized to ensure good stability of the amplifier. Refer to the seed laser user's manual. It is important that the duration of the seed pulses not be too long, because the stretcher may not sufficiently reduce their peak power to avoid damage to the amplifier. Use a scanning autocorrelator to monitor the seed pulse duration.

Seed Beam Alignment into the Regenerative Amplifier

Figure 6-2 shows the components used to align the seed beam into the amplifier. A description of the *Spitfire Pro* design is given in Chapter 7.

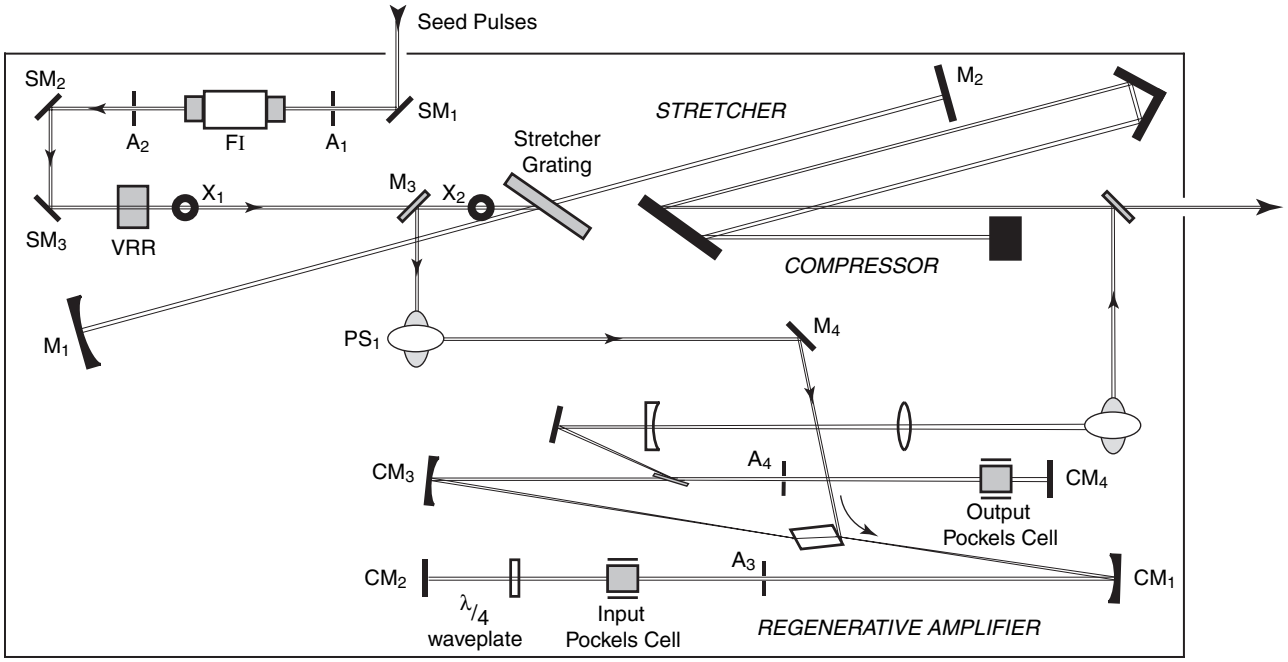


Figure 6-2: Optical Path for Seed Beam Alignment

The input beam is directed by SM_1 through Faraday Isolator FI and the first two alignment apertures, A_1 and A_2. It is then routed by seed mirrors SM_2 and SM_3 through the vertical retro-reflector VRR and onto the stretcher diffraction grating. SM_1, SM_2 and SM_3 all have vertical and horizontal adjustments.

1. Verify the pump beam shuttered is closed.
2. Check the alignment of the seed beam through apertures A_1 and A_2 and make small adjustments as necessary.
3. Install the removable reference iris in the X_1 location (see Figure 6-2) and adjust the aperture opening to about 4 mm diameter.
4. Adjust mirror SM_2 to center the seed beam through the aperture of the removable iris.
5. Relocate the removable iris to the new location X_2 shown in Figure 6-2.
6. Adjust mirror SM_3 to center the cavity-dumped beam through the aperture of the removable iris.
7. Return the aperture to the X_1 location and repeat Step 6. Iterate Steps 6 through 9 until the beam is aligned through the aperture in both locations with no further adjustments needed to either SM_2 or SM_3.

   When correctly aligned, the 1st order diffracted beam from the stretcher grating will strike the center of gold mirror M_1.

8. Check the alignment of the seed beam into the amplifier.

   It is possible that it will have drifted slightly since the *Spitfire Pro* was last operated. The beam should be aligned using PS_1 and M_4 so that it is centered on the input Pockels cell and then onto cavity mirror CM_2.

9. Use the IR card to make slight adjustments to mirror M_4, *not the cavity mirrors*, until you see a beam at CM_4.

   Enough of the beam should pass back through the input Pockels cell, the Ti:sapphire rod, and the other components in the optical path so that it is visible on the IR card in front of cavity mirror CM_4.

Beam Uniformity

The *Spitfire Pro* amplifier is designed to produce a near Gaussian output beam. Beam uniformity is easily checked by visually inspecting the pattern on a white business card.

A poor beam is often an indication of misalignment of the pump beam. Refer to Appendix A for procedures to optimize pump beam alignment (refer to Chapter 7 for a description of the pump beam path.)

Caution!

Make only small and reversible changes to the pump beam alignment. The pump beam is tightly focused in the Ti:sapphire rod in the amplifier, and is easy to misalign. Refer to Appendix A for a description of the pump beam alignment procedure. It is recommended that you contact your Spectra-Physics representative before making adjustments to the pump laser.

Optimizing the Regenerative Amplifier

Use this procedure to minimize the time it takes for the amplified pulse to reach its peak level. This "build-up time" is compared to the time it takes for the *Spitfire Pro* to amplify its own spontaneous emission in the absence of a seed pulse. Minimizing the time for the pulse to be amplified is called "build-up time reduction."

Note

Optimizing the build-up time reduction is fundamental to optimizing the performance of the *Spitfire Pro*.

Danger!
Laser Radiation

In the following procedure, the *Spitfire Pro* is initially operated as an optically-pumped, Q-switched laser that is capable of producing >1.5 W of average power at a wavelength near 800 nm. Use appropriate caution.

1. Block the seed beam (where it leaves the stretcher) from entering the regenerative amplifier.
2. Open the pump laser shutter.
3. Disable the output pockels cell by clicking on the button below PC2 on the Main menu (it turns gray).

The regenerative amplifier will begin to operate as a laser. Allow it to stabilize for about 5 minutes.

4. Monitor the intracavity pulse using the output of the photodiode behind CM_4.

   The location of the BNC connector for this output is shown in Figure 6-3. Use a fast oscilloscope with a micro-channel plate screen, or a digitizing oscilloscope with a sampling rate greater than 2 Giga-samples/sec.

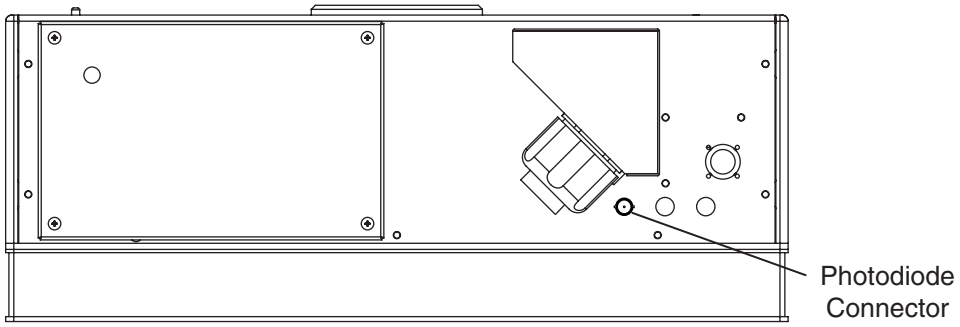


Figure 6-3: Location of Photodiode Connector on the *Spitfire Pro*

Trigger the oscilloscope externally using the *TDG* SYNC OUTPUT connector. Use a 50 Ω input impedance for the photodiode, and set the time-base to 100 or 200 ns/div. The pulse should look like the one shown in Figure 6-4.

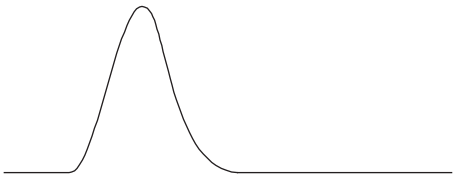


Figure 6-4: Appearance of Q-switched Pulse

5. Unblock the seed beam.

   The energy of the seed laser pulses will exceed the energy of the spontaneous emission in the Ti:sapphire rod in the regenerative amplifier. As a result, the rod will now preferentially amplify the seed pulses. The intracavity radiation should look like that shown in Figure 6-5.

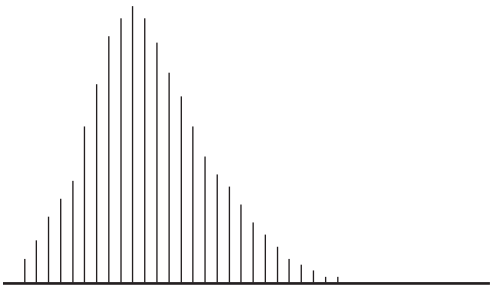


Figure 6-5: Intracavity Pulse Train

6. *Reduce the pulse build-up time to the minimum possible.*

   While monitoring the pulse train, make small, iterative adjustments to turning mirrors PS_1 and M_4 (which direct the seed pulse into the regenerative amplifier) so that the pulse train moves to the left on the oscilloscope screen.

7. Next, enable and adjust PC2 until the intracavity pulse train looks like that shown in Figure 6-6.

   Note that the most stable performance is obtained by adjusting the timing so that the pulse train includes the one pulse that is just past the maximum.

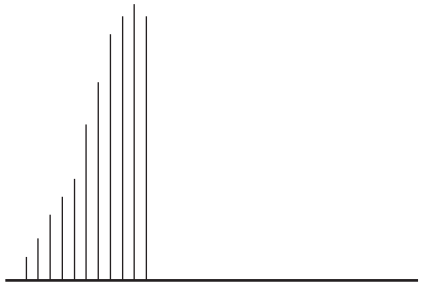


Figure 6-6: Intracavity Pulse Train with the Timing Set Correctly

8. Set up a fast photodiode to sample the output of the *Spitfire Pro* and view the output pulse on the digitizing oscilloscope. (Use the same settings given in Step 4). A single, stable output pulse should be displayed.
9. Observe the output mode on a white business card. If adjustment is necessary, refer to the pump beam alignment procedure given in Appendix A.
10. If the pulse amplitude is stable but there is evidence of a secondary pulse, make a slight adjustment to the PC2 control. If this does not produce a single, stable, cavity-dumped pulse, adjust PC2 ± 10 ns.
11. Verify the period between the pulses represented by Figure 6-5 is about 10 ns, which indicates that only integral pulses are being injected into the amplifier.

    Pulses separated by about 5 ns, for example, are a indication of a timing problem. Make slight adjustments if necessary to PC1 until the pulse train looks like that shown in Figure 6-5.

If this procedure does not produce a single, stable, cavity-dumped pulse, contact your Spectra-Physics service engineer.

The Spitfire Pro GUI Control Software

Note

The *Spitfire Pro* GUI provides software control of the *Spitfire Pro* amplifier and the *Empower* pump laser. It does *not* control the operation of either a *Tsunami* or *Mai Tai* seed laser. Refer to the seed laser user's manual for instructions for operating the seed laser.

Main Menu

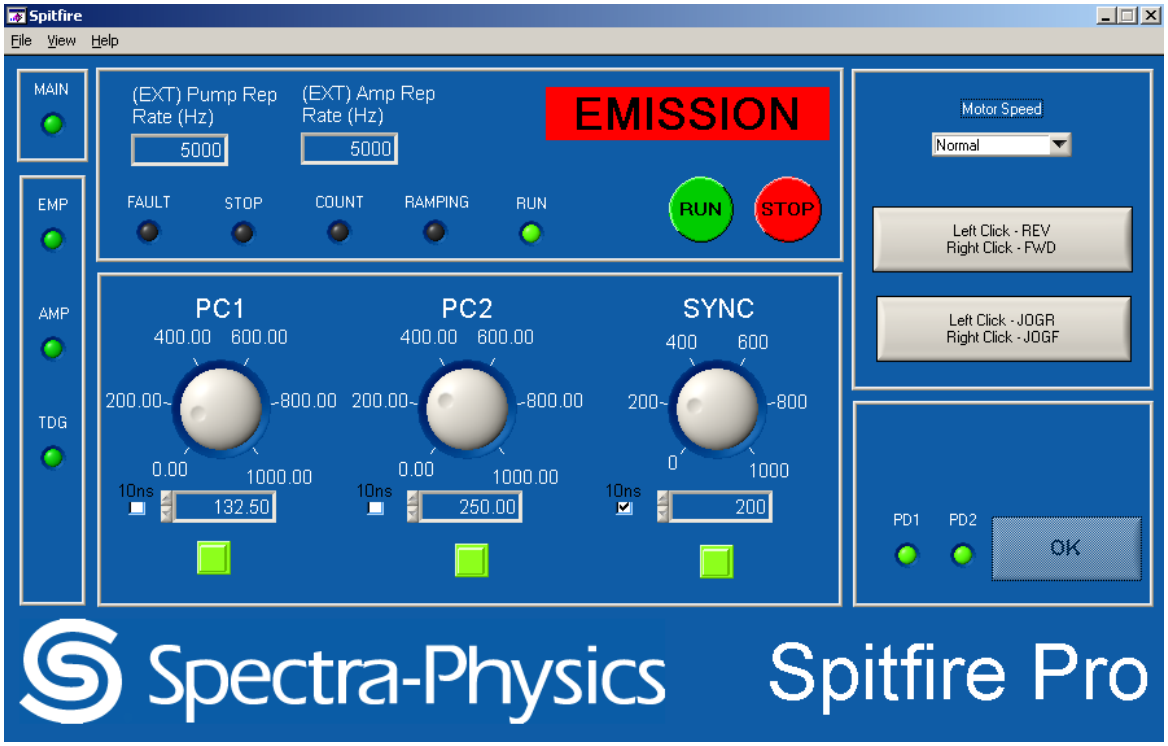


Figure 6-7: The Main Menu

Communication status indicators—the column of four indicators on the far left of the Main menu report that the health of the following communications links is good when they are green:

MAIN indicator—when on, indicates the Ethernet communication link between the computer and the *TDG* is operating properly.

EMP indicator—when on, indicates the communication link between the *Empower* power supply and the *TDG* is operating properly.

AMP indicator—when on, indicates the communication link between the *Spitfire Pro* and the *TDG* is operating properly.

TDG indicator—when on, indicates the communications within the *TDG* itself is operating properly.

If one of the communication links is lost (indicator turns gray), exit the software, check the cable connections and restart the software.

(EXT) PUMP REP RATE (HZ)—displays the pump repetition rate, in Hertz.

(EXT) AMP REP RATE (HZ)—displays the amplifier repetition rate, in Hertz.

FAULT indicator—illuminates if a fault (e.g., an open interlock) is detected. See the "Active Faults Menu" on page 6-13 for more information.

RUN/STOP buttons—turn the *Empower* pump laser on and off.

Clicking and holding the RUN button for about 5 seconds turns on the diode laser arrays in the *Empower* laser. The red EMISSION bar also appears on the display and, after another brief delay (during which time the COUNT indicator illuminates and the system beeps), the shutter on the *Empower* laser head opens with an audible click. The white emission indicator on the *Empower* laser head flashes and the COUNT indicator turns off.

When the *Empower* lases, it will be at reduced power. The RAMPING indicator illuminates on the Main menu to show that the diode laser arrays have begun a controlled ramp-up in power to allow time for the doubling crystal to maintain operating temperature. When the sequence is complete and the diode pump lasers have reached their pre-set current, the RAMPING indicator turns off.

When STOP is pressed, the *Empower* stops lasing (the pump laser diodes turn off and the internal shutter in the laser cavity closes).

PC1, PC2, SYNC Controls:

PC1 corresponds to the CH1 connector on the back panel of the *TDG* and is used to control the timing of the input Pockels cell.

PC2 corresponds to the CH2 connector on the back panel of the *TDG* and is used to control the timing of the output Pockels cell.

SYNC corresponds to the SYNC OUTPUT connector on the front panel of the *TDG* and allows the user to synchronize laboratory equipment with the amplified output pulse.

Dial control—is used to select a value between 0 and 1000 ns, which is displayed in the data box below it. Alternatively, the up-down buttons are used to adjust the value, and the display and dial will reflect the adjusted value.

10NS check box—sets the increment size to 10 ns when checked and displayed values are shown without decimal places. An unchecked box sets the increment size at 0.25 ns and displayed values are shown with 2 decimal places. Use an unchecked box for fine adjustments or a checked box for faster adjustments.

Push buttons—toggle between green (on) and gray (off) to activate or de-activate the Pockels cells.

Compression Motor Controls:

MOTOR SPEED window—is used to set the speed of the motor that controls the optical length of the compressor to fast, normal, slow or very slow motion.

REV/FWD button—lengthens the path length in the compressor as long as the left mouse button is held down and shortens the path length in the compressor as long as the right mouse button is held down.

JOGR/JOGF button—lengthens the compressor path slightly with a left mouse click; a right click shortens the compressor path slightly.

BWD controls

PD1, PD2 indicators—when both are on, indicate that the stretcher is spreading the seed pulse spectrum properly on the tall stretcher end mirror. PD_1 represents the red end of the spectrum; PD_2 represents the blue end. If an indicator is off, the corresponding photodetector is receiving a signal below threshold and amplifier output is turned off.

RESET/OK button—resets the relay when the button is pressed while it shows "RESET." *Spitfire Pro* amplification resumes after the underlying problem is resolved and both BWD indicators are green. When this is the case, the button shows "OK."

Clicking "View" on the menu bar activates a pull-down list offering a choice of three additional displays:

- Empower Setup
- Timing and High Voltage
- Active Faults

The Main menu should remain visible when one of the above displays is selected. Click in its window (or close the other windows) to return to the Main menu. Each of these additional displays is described below.

Empower Setup Menu

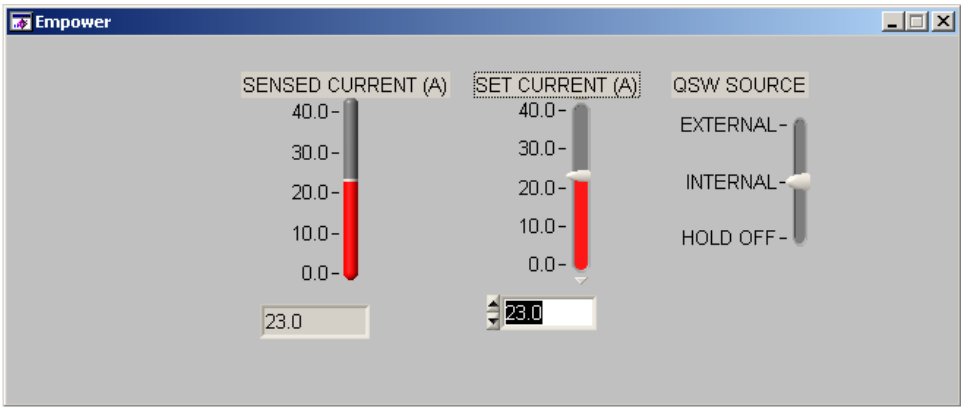


Figure 6-8: The Spitfire GUI Empower Setup Menu

QSW SOURCE slide control—sets the Q-switch trigger source:

EXTERNAL—selects the EXT QS TRIG IN BNC on the back of the power supply as the Q-switch trigger source. The repetition rate is pre-set at the factory to the value specified when the system was ordered. The repetition rate using EXTERNAL control can be varied ±20% from the pre-set value (e.g., from 800 Hz to 1.2 kHz for a 1 kHz system). Repetition rates cannot be set less than 50 Hz or more than 10 kHz.

The external trigger frequency must be greater than 50 Hz or an interlock fault will result, turning off the laser. RF power for the Q-switches is sent to the laser head when this mode is selected.

INTERNAL—the Q-switch trigger source is the internal clock in the power supply. The repetition rate is pre-set at the factory to the value specified when the system was ordered. RF power for the Q-switches is sent to the laser head when this mode is selected.

Using INTERNAL control to change the repetition rate from the value pre-set at the factory requires a service call to Spectra-Physics.

HOLD OFF—this mode is only used during service procedures.

Note

Changing from HOLD OFF to INTERNAL or EXTERNAL while the laser is running will cause the laser to stop. To restart the laser, click on RUN.

SET CURRENT [A] control—is used to set the current (in amps) for the diode pump lasers, up to the factory maximum set point. The numeric value can be entered directly or in 0.1 A steps using the slide control.

SENSED CURRENT [A]—displays actual diode laser current.

Timing and High Voltage Menu

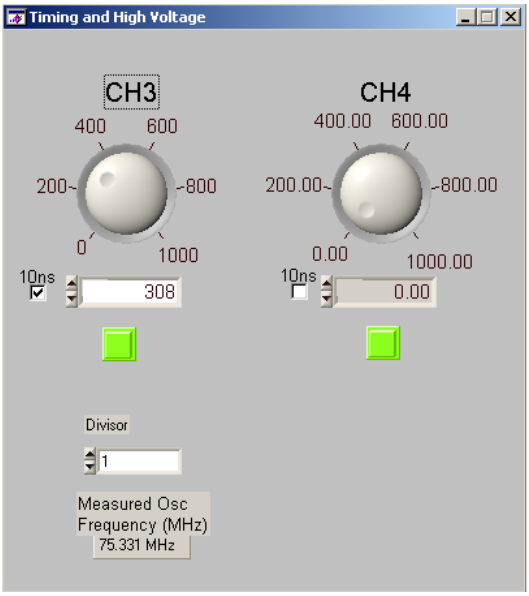


Figure 6-9: The Spitfire GUI Timing and High Voltage Menu

CH3, CH4 controls—can be used for additional output timing signals for synchronization of laboratory equipment.

Dial control—is used to select a value between 0 and 1000 ns, which is displayed in the data box below it. Alternatively, the up-down buttons are used to adjust the value, and the display and dial will reflect the adjusted value.

10NS check box —when checked, sets the increment size to 10 ns and the displayed values are shown without decimal places. An unchecked box sets the increment size at 0.25 ns and the displayed values are shown with 2 decimal places. Use an unchecked box for fine adjustments or a checked box for faster adjustments.

Push buttons—toggle between green and gray when clicked to activate or de-activate these outputs.

DIVISOR data entry box—accepts integer divisors for reducing the basic repetition rate of the *Spitfire Pro* pulsed output (e.g., a divisor of 2 will decrease the output of a 1 kHz system to 500 Hz).

MEASURED OSC FREQUENCY—displays the measured oscillation frequency in Megahertz.

Active Faults Menu

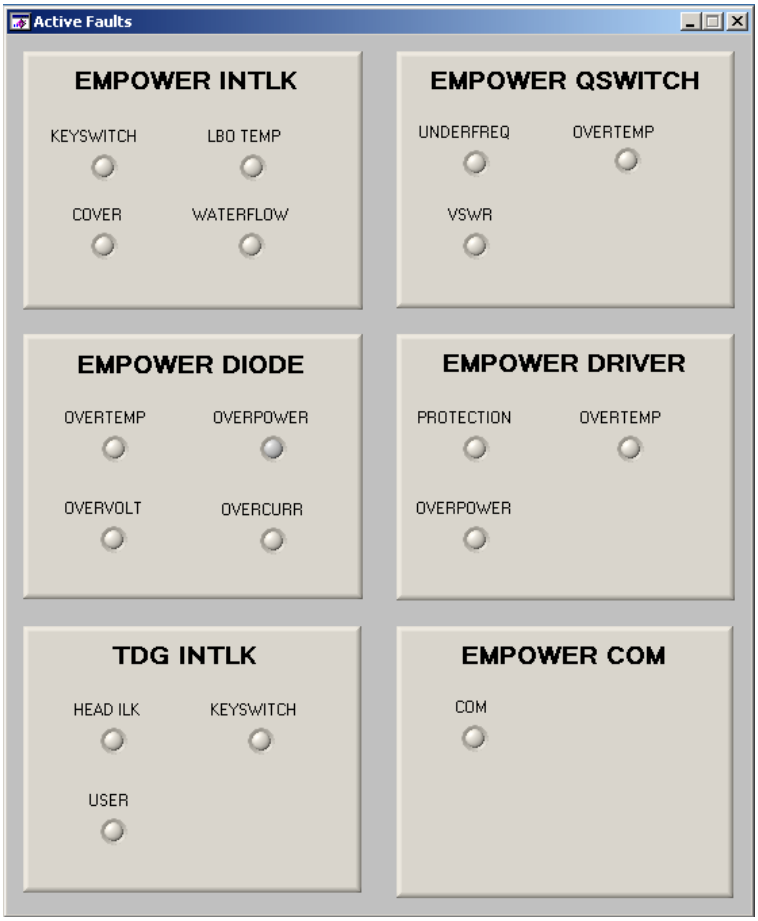


Figure 6-10: The Spitfire GUI Empower Active Faults Display

This menu represents a total of 18 possible system faults arranged in 5 categories for the *Empower* pump laser and 1 category for the *TDG*. An indicator is on (green) if the corresponding fault is presently active (i.e., a fault exists). All indicators must be off (gray) to enable normal operation.

EMPOWER INTLK indicators—turn on (are green) when one of the following interlocks is detected as open:

KEYSWITCH indicator—turns on if the *Empower* power supply keyswitch is in the OFF position.

LBO TEMP indicator—turns on when the LBO temperature is outside the factory set points. Allow the temperature to return to the specified temperature range before turning on the system again.

COVER indicator—turns on when the *Empower* power supply user interlock is open. Verify that either the interlock switch is closed or the interlock jumper plug is in place. Cycle the *Empower* power supply keyswitch to clear the fault.

(This indicator also turns on if any of the micro-switches inside the internal laser head sense that the internal cover has been opened. This

can seriously compromise the cleanliness of the laser cavity. Contact your Spectra-Physics representative.)

WATERFLOW indicator—turns on when the *Empower* chiller water flow is less than 2.0 GPM. Verify the chiller is functioning properly and, to ensure adequate flow, that the water pressure and reservoir level are correct and that the water hoses are not kinked. Cycle the *Empower* power supply keyswitch to clear the fault.

EMPOWER QSWITCH indicators—turn on when one of the following Q-switch fault conditions is detected:

UNDERFREQ indicator—turns on when QSW MODE is set to EXTERNAL and the external source of triggering drops below 50 Hz. Ensure that the external Q-switch triggering source is producing TTL pulses at 50 Hz or higher.

OVERTEMP indicator—turns on when the thermal sensor in either AOM indicates the safe operating temperature for that device is exceeded. Verify the chiller is functioning properly and, to ensure adequate flow, that the water pressure and reservoir level are correct and that the water hoses are not kinked. Cycle the *Empower* power supply keyswitch to clear the fault.

VSWR indicator—turns on when either or both of the QSW RF connectors on the back of the *Empower* power supply are disconnected, or when some connection fault is causing poor coupling of RF power into the AOMs.

EMPOWER DIODE indicators—turn on when a fault condition is detected in the diode laser pump chamber inside the *Empower* laser head. This would indicate a serious condition capable of causing permanent damage to the *Empower* if not quickly corrected.

OVERTEMP indicator—turns on when the temperature sensor in the pump chamber exceeds the safe operating temperature. Turn off the *Empower* power supply and allow the chiller to run for 10 minutes to cool it down. Then verify proper chiller temperature and restart the laser, running it at a lower diode laser drive current.

OVERPOWER indicator—turns on when diode pump laser power exceeds a factory-set limit. Reduce the diode laser drive current, then cycle the *Empower* power supply keyswitch to clear the fault.

OVERVOLT indicator—turns on when the voltage required to supply the requested current to the diode lasers exceeds the factory-set limit. Check the pump chamber umbilical connection, then cycle the *Empower* power supply keyswitch to clear the fault.

OVERCURR indicator—turns on when the current requested to drive the diode laser array exceeds the factory-set limit. Lower the current drive setting, then cycle the *Empower* power supply keyswitch to clear the fault.

EMPOWER DRIVER indicators—turn on when a fault condition is detected in the diode lasers drive electronics. This indicates a serious condition capable of causing permanent damage to the *Empower* if not quickly corrected.

PROTECTION indicator—turns on when the internal drive FET protection circuit is activated.

Warning!



Do not operate the *Empower* if the PROTECTION fault occurs! Contact your Spectra-Physics service representative.

OVERTEMP indicator—turns on when the temperature of the diode laser driver (FET) heatsink becomes overheated due to inadequate airflow to the driver, or to a shorted or over-current condition. Verify that airflow in the power supply is adequate and allow the system to run without lasing for at least 10 minutes to cool it down, then restart the laser.

OVERPOWER indicator—turns on when the internal drive FET exceeds a factory-set power limit. Reduce the diode drive current, then cycle the *Empower* power supply keyswitch to clear the fault.

EMPOWER COM indicator—turns on when communications are interrupted between the computer and the power supply.

COM indicator—turns on when any communication interruption is detected. Check the PWR SUPPLY COMM connections and verify that the power supply is turned on. Exit the *Spitfire* control software, cycle the *Empower* power supply keyswitch, then restart the *Spitfire* software to clear the fault.

Chapter 7 The Spitfire Pro Beam Path

The beam path and its adjustment through the *Spitfire Pro* are described below to assist in making adjustments to the optical components.

Note

When describing the beam path, "left" and "right" refer to beam adjustment direction as seen moving along the beam from input to output.

A few of the more complex optical elements require some initial description. Refer to Figure 7-1.

Faraday Isolator (FI)—protects the seed laser components by absorbing any reflected power generated in the amplifier and pulses that are not selected for amplification. There are no adjustments for this device.

Gold Mirror (M_1)—directs and collimates the beam dispersed by the stretcher grating onto the Tall Stretcher Mirror (M_1), then directs the return beam back again to the stretcher grating. It has vertical and horizontal adjustments.

Tall Stretcher End Mirror (M_2)—reflects the beam back onto the Gold Mirror (M_1). It has vertical and horizontal adjustments.

Bandwidth Detector (BWD) (not shown in the drawings)—is a safety device that protects the system when there is not enough bandwidth in the seed pulse for it to be properly spread by the stretcher (usually caused by a misaligned or poorly mode-locked seed laser). See Chapter 4, "Controls, Indicators and Connections," for more information about the BWD. This device is pre-set at the factory.

Vertical Retroreflectors (x2)—comprise a pair of flat mirrors at right angles that translate the beam up or down and reflects it back on a parallel path. There is an assembly in the stretcher and one in the compressor; each has vertical and horizontal adjustments that are set at the factory and must not be disturbed.

Horizontal Retroreflector—translates the beam sideways and reflects it back on a parallel path. This assembly has vertical and horizontal adjustments that are set at the factory and must not be disturbed. It is also mounted on a translational track that is moved using the MOTOR CONTROL function on the Main menu to optimize compressor length for the shortest duration pulses possible.

Polarizer—is an optical element that is transparent to horizontally polarized light and reflects (rejects) vertically polarized light. It is used to direct amplified pulses into the compressor. Mounting screws provide vertical and lateral movement for alignment.

Stretcher and Compressor Beam Paths

Although the beam passes from the stretcher into the amplifier and then to the compressor, the beam path in the stretcher and in the compressor are described here together since they share the same compartment and their treatment of the beam is similar.

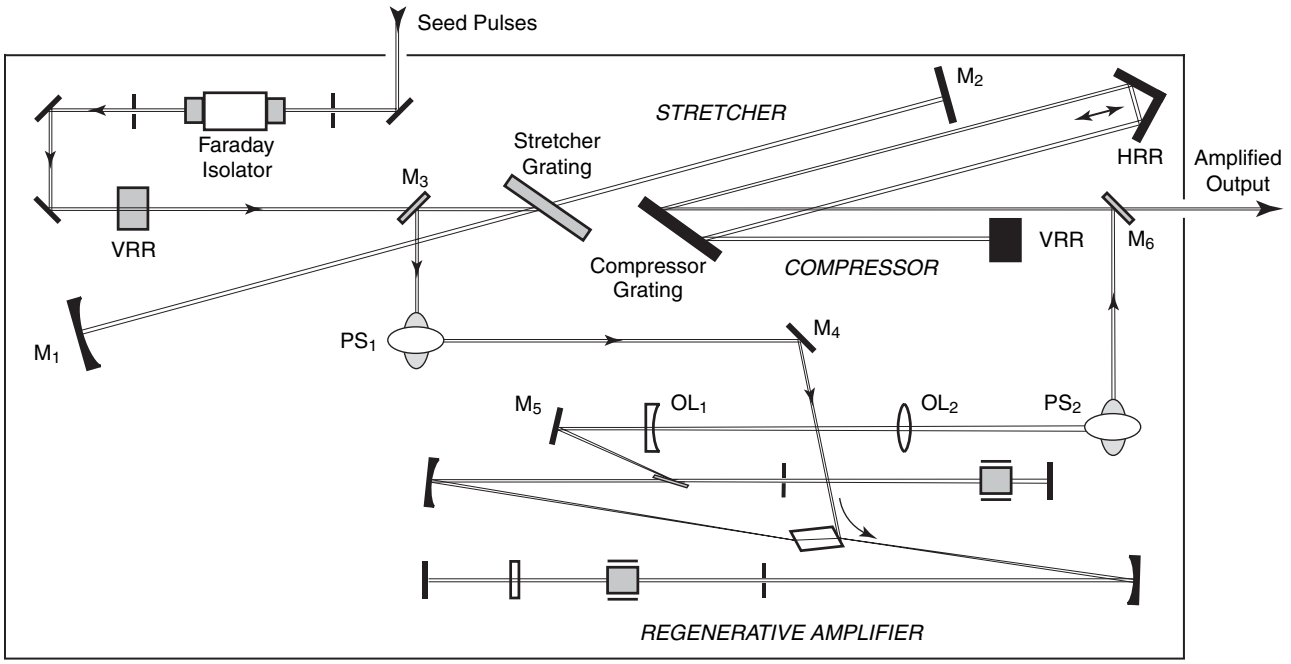


Figure 7-1: Optical Components in the *Spitfire Pro F* (Stretcher and Compressor)

The different versions of the *Spitfire Pro* have different stretcher and compressor designs because of their different pulse widths. Basically, each version uses its own set of gratings that are set at different angles to the beam. There are also other minor differences.

Although your version of the *Spitfire Pro* may differ slightly, study the beam path through the *Spitfire Pro F* model shown in Figure 7-1. Differences in stretcher/compressor design of other versions are described relative to this design in later sections.

After passing through the Faraday isolator, the seed laser beam is horizontally polarized and remains so in the stretcher. The polarization of the beam is changed to vertical in the amplifier, but is again horizontally polarized when it enters the compressor.

Numbers are used in the following drawings to track the beam as it passes from optic to optic. The numbers are not used either to name the optic itself or to indicate the position of the beam. Refer to Figure 7-1 for the abbreviations used to name components in the stretcher and compressor.

Note

Shorter and longer wavelengths strike the optical components in the stretcher and compressor at different locations. Figure 7-2 shows how the wavelengths are separated.

The Spitfire Pro F Stretcher

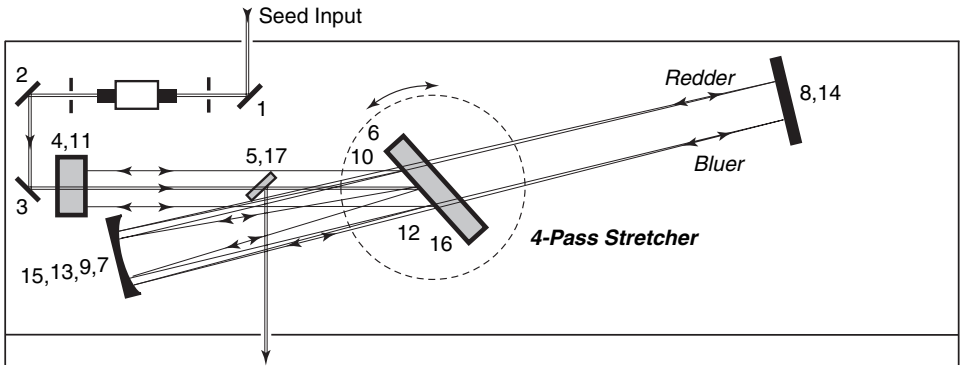


Figure 7-2: *Spitfire Pro F* Stretcher Beam Path

The vertically polarized seed beam is first routed through the Faraday isolator (1) before entering the stretcher (2, 3).

1. To enter the stretcher, the seed beam passes *through* a gap in the vertical retroreflector VRR (4) and *over* the pick-off mirror, M_3 (5).
2. The grating spreads the beam spectrally (6) and directs the broadening beam onto the center of the concave gold mirror, M_1 (7). The grating mount has a single adjustment that rotates the grating to change the angle of incidence of the beam.
3. M_1 is angled slightly upward to reflect the beam *over* the grating (8) and onto the tall stretcher end mirror, M_2. The concave gold mirror and the tall stretcher end mirror have vertical and horizontal adjustments.
4. M_2 reflects the beam back *over* the grating to M_1 (9), which returns it to the grating (10).
5. The grating reflects the collimated (and diffracted) beam toward the bottom of the vertical retroreflector VRR (11). Notice that the path of the redder wavelengths is longer than that of the bluer wavelengths and, therefore, lags behind the bluer wavelengths.
6. The beam now retraces its path back through the stretcher. VRR (12) reflects the beam back to the top of the grating. The spectrum is temporally spread even further as the redder wavelengths again take the longer path. Passing the beam through the stretcher one more time (13, 14, 15, 16), it is focused back into a round beam. However, the bluer components are now well ahead of the red.
7. Because the beam hits high on the concave mirror (15), it is reflected to the bottom of the grating (16), and as it leaves the grating it is now low enough to be picked off by mirror M_3 (17). It exits the stretcher and is routed into the regenerative amplifier. M_3 has vertical and horizontal adjustments.

The Spitfire Pro F Compressor

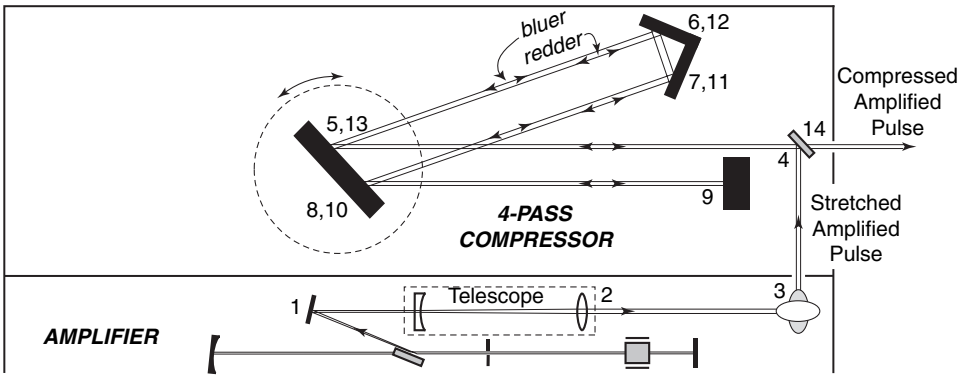


Figure 7-3: *Spitfire Pro F* Compressor Beam Path

The temporally stretched, pulsed beam passes from the stretcher into the regenerative amplifier. After it achieves its maximum level of amplification, the beam is then ejected out of the regenerative amplifier by the horizontal polarizer (see the discussion following Figure 7-7 and Figure 7-8 for more details). The mechanism for ejecting the beam from the amplifier is discussed in "The Ti:Sapphire Regenerative Amplifier" on page 7-7.

1. The beam reflects from M_5 (1) which directs the vertically polarized beam through the expanding telescope (2) (comprising OL_1 and OL_2) in order to reduce the beam intensity in the compressor.
2. Polarizing periscope PS_2 (3) rotates the beam to horizontal polarization and directs it to the compressor routing mirror M_6 (4), which then sends it onto the right side of the compressor grating (5). M_6 has vertical and horizontal adjustments.
3. The grating spreads and reflects the beam towards the horizontal retroreflector HRR (6, 7) with the redder wavelengths on the right and the bluer wavelengths on the left. The grating mount has a rotational adjustment.
4. The horizontal HHR steps the beam over about two inches, flips the ends of the spectrum, and returns the beam to the lower left side of the grating (8). The redder wavelengths now take the shorter path.
5. The beam is reflected by the grating and impinges on the VRR (9) where it is stepped upwards an inch and is sent back to the top left side of the grating (10).
6. The grating begins to refocus the beam and reflects it to the horizontal retroreflector (11, 12).
7. The horizontal retroreflector flips the beam around again and sends it back to the grating (13) where the beam is compressed back close to its original duration.
8. The beam is reflected *over* M_6 (14) and exits the *Spitfire Pro*.

Spitfire Pro PM Stretcher and Compressor

The picomask version of the *Spitfire Pro* is identical to the *Spitfire Pro F*, except that it uses gratings designed for a picosecond seed laser, and a special mask aperture is added to the stretcher cavity. This picomask clips the seed beam, thereby reducing the bandwidth of the femtosecond pulses by about a factor of ten so that they have the equivalent bandwidth of a picosecond seed laser.

As noted earlier, arbitrarily clipping the seed beam will result in damage to the optical components of the amplifier if the stretcher grating fails to adequately reduce the peak power of the input seed pulses. The *Spitfire Pro PM* combines a precision mask with a stretcher grating that uses a high ruling density designed to disperse the reduced bandwidth pulses.

The *Spitfire Pro PM* does not require the longer path length in the stretcher and compressor that are required by an amplifier seeded by a picosecond laser. The picomask and its position in the stretcher in front of M_1 are shown in the following figures.

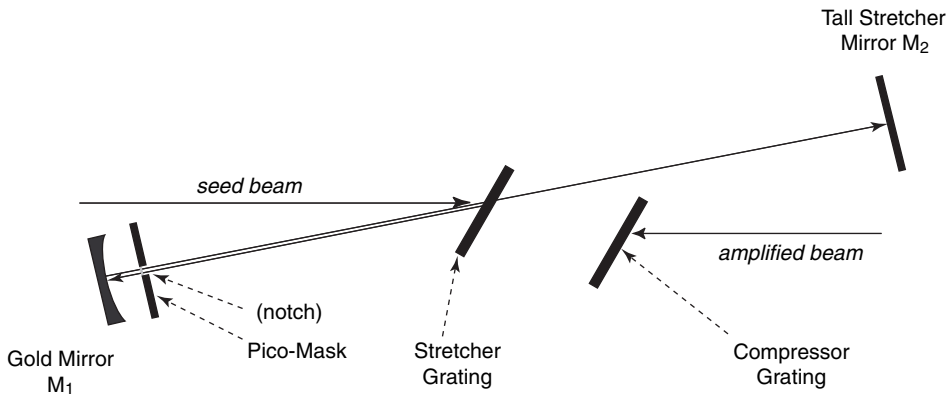


Figure 7-4: *Spitfire Pro PM* Picomask Location

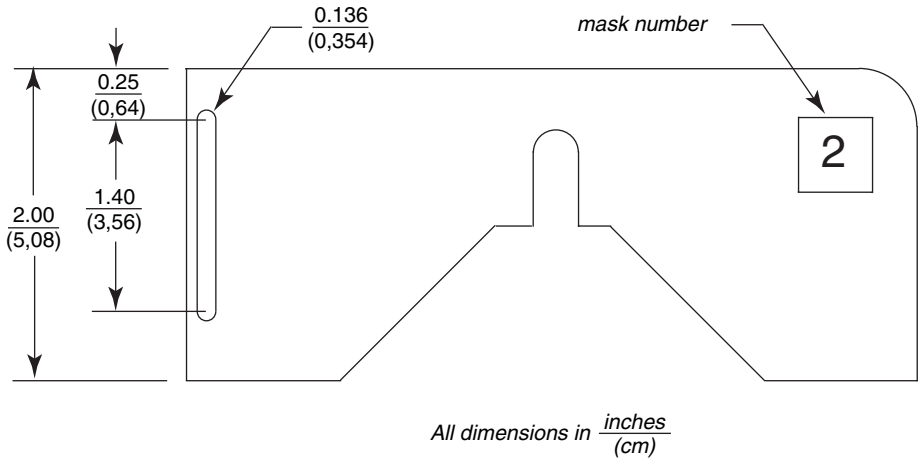


Figure 7-5: *Spitfire Pro PM* Picomask

Spitfire Pro 40F Stretcher and Compressor

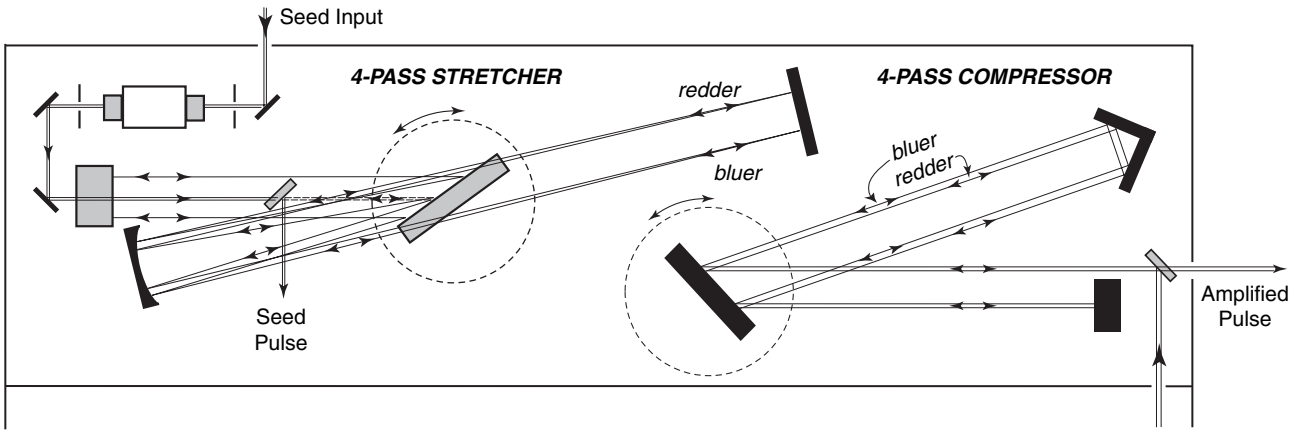


Figure 7-6: *Spitfire Pro 40F* Stretcher and Compressor Beam Path

Amplification of pulses of the shortest duration requires that extra attention be paid to correcting the dispersion that occurs within the amplifier cavity.

Each Fourier frequency component of a pulse experiences a slightly different index of refraction as it propagates though a material, causing a time delay between the different frequencies. Group Velocity Dispersion (GVD) is defined as the variation in the time delay of a pulse as a function of wavelength. Typically, GVD causes red frequencies to travel faster than blue frequencies. The effect is more pronounced for shorter pulses, such as those amplified by the *Spitfire Pro 40F*.

In addition to GVD, the pulse width is affected by the nonlinear index of Ti:sapphire, which results in self phase modulation (SPM). As the pulse propagates through the Ti:sapphire material, the leading edge is "red-shifted" by an increasing index of refraction. Conversely, the trailing edge of the pulse is "blue-shifted." (More information about GVD, SPM, and dispersion compensation can be found in the *Mai Tai* or *Tsunami* user's manuals.)

In order to achieve near transform-limited output pulses, it is necessary to compensate for the pulse spreading caused by positive GVD and SPM. This is accomplished by using a compressor grating with a higher ruling density than the stretcher grating. Doing so, however, makes this design different from that used in other *Spitfire Pro* models, which use the same grating density for stretcher and compressor. Furthermore, using a different ruling density for each grating requires each grating to be presented to the beam at a different angle.

However, the same four-pass design can be used for the beam paths in the stretcher and in the compressor in order to obtain adequate spatial separation of the pulse's wavelength components. The beam in the *Spitfire Pro 40F* follows the same sequence from optic to optic as outlined earlier for the *Spitfire Pro F*.

The Ti:Sapphire Regenerative Amplifier

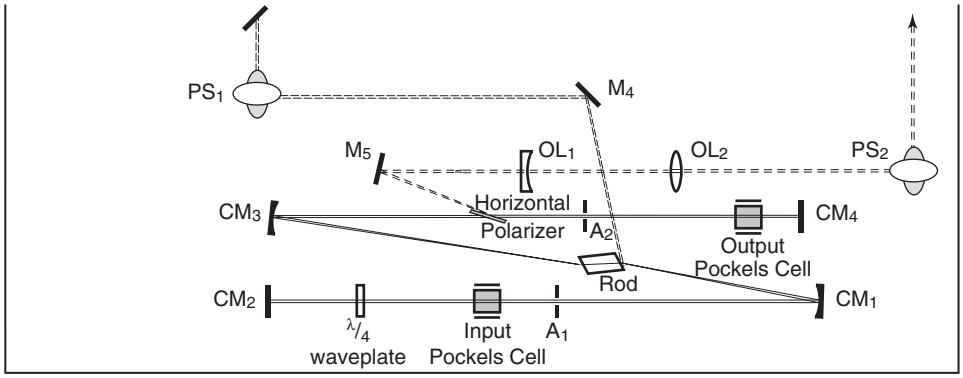


Figure 7-7: Regenerative Amplifier Optical Components

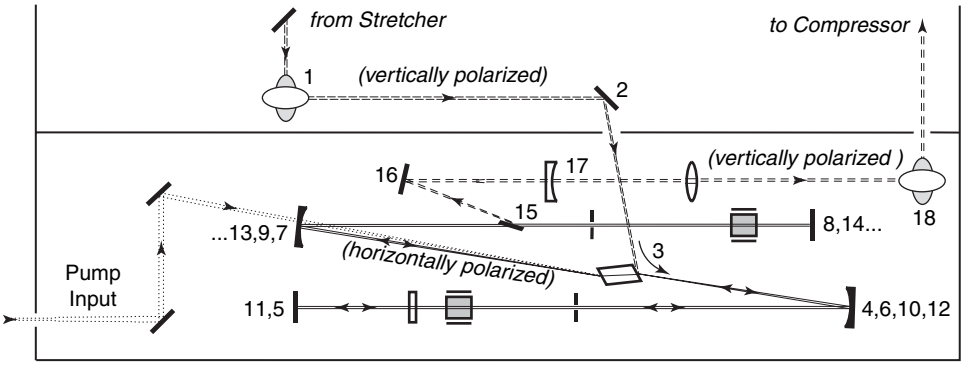


Figure 7-8: Regenerative Amplifier Beam Path

Spitfire Pro models all make use of a regenerative amplifier in a Z-shaped folded cavity. Refer to Figure 7-7 for component names (M_1, $\lambda/4$ etc.). The beam path is shown in Figure 7-8. For clarity, some components, such as apertures and lenses, are not shown.

The Seed Beam Path

1. Horizontally polarized pulses from the stretcher are rotated (1) to vertical polarization by the polarization rotating periscope PS_1 and are directed into the amplifier cavity by mirror M_4 (2).
2. The Ti:sapphire rod, cut at Brewster's angle for horizontally polarized light, reflects the vertically polarized pulses off its surface (3) and directs them to the first cavity mirror CM_1 (4), which directs the pulses to the input Pockels cell.

At this point, the pulsed beam is in the amplifier cavity.

Whether a particular pulse remains in the cavity to be amplified is determined by the input Pockels cell. When this Pockels cell is off, it is transparent to both vertically polarized and horizontally polarized light. When the input Pockels cell is on, combined with the ¼ waveplate ($\lambda/4$), it rotates the polarization of the beam 90°.

Pulse selection is accomplished by using the polarization rotating properties of the passive $\lambda/4$ together with the input Pockels cell. Pulses at kilohertz rates are selected for amplification while the remaining megahertz seed pulses are rejected. Control of pulse selection is determined by the *TDG*, as described in Chapter 4.

Note

As long as the selected pulse remains horizontally polarized, it remains in the cavity. Whenever a pulse arrives at the Ti:sapphire crystal as vertically polarized, it is reflected off the surface and is not amplified.

One of three things will now happen:

Case (a): the input Pockels cell is off and stays off (pulse is rejected)

1. The vertically polarized pulse reflects off CM_1 (4), passes through aperture A_1, through the inactive input Pockels cell, and is rotated 45° as it passes through $\lambda/4$. It reflects off CM_2 (5) and rotates another 45° as it passes through $\lambda/4$ again.
2. The pulse, now horizontally polarized, passes through the inactive input Pockels cell again, through the aperture, and is reflected by CM_1 (6), this time to the Ti:sapphire rod. Because it is now horizontally polarized, it passes through the rod and picks up first-pass gain.
3. The pulse is reflected by CM_3 (7) through the horizontal polarizer, through aperture A_2 and through the inactive output Pockels cell.
4. The pulse reflects off CM_4 (8) and passes back through the inactive output Pockels cell, through aperture A_2, through the horizontal polarizer, reflects off CM_3 (9), and passes back through the Ti:sapphire rod for second pass gain.
5. The beam reflects from CM_1 (10), passes through the inactive input Pockels cell, and is again reflected back from CM_2 (11). Having passed twice through $\lambda/4$, it is now vertically polarized and is reflected from the surface of the Ti:sapphire rod and out of the amplifier cavity to M_4. Once rejected, the pulse passes back through the stretcher and is absorbed by the Faraday isolator.

Case (b): Input Pockels cell is already on (pulse is rejected)

1. The incoming vertically polarized pulse reflects off CM_1 (4) and passes through aperture A_1. But this time, as it passes through the now active input Pockels cell, it is rotated 45° by the cell and another 45° as it passes through $\lambda/4$, becoming horizontally polarized. After reflecting off CM_2 (5), it is rotated another 45° by $\lambda/4$ and another 45° by the still active input Pockels cell, and returns to vertical polarization. It passes through aperture A_1, reflects off CM_1 (6) and is reflected off the Ti:sapphire surface and ejected out of the cavity.

Case (c): Input Pockels cell is off and then is turned on (pulse is selected)

1. The soon-to-be-selected, vertically polarized pulse reflects off CM_1 (4), passes through aperture A_1, through the inactive input Pockels cell, and is rotated 45° as it passes through $\lambda/4$. It reflects off CM_2 (5) and rotates another 45° as it passes through $\lambda/4$ again.

This time, *after* the pulse passes back through the inactive Pockels cell and travels toward CM_1 (6), the input Pockels cell is turned on. (The output Pockels cell remains off for now.)

The pulse remains in the cavity because it remains horizontally polarized. (Since the input Pockels cell is on, the pulse is flipped 180° each time it traverses the input path, leaving its polarization unchanged.) The pulse is then amplified each time it passes through the crystal. The pulses that follow behind the selected pulse arrive with the input Pockels cell already turned on. These following pulses remain vertically polarized as in Case (b), and are discarded.

2. After the selected pulse has passed through the crystal about 20−25 times (6 through 14...), it has reached its optimum amplification. The output Pockels cell is now turned on just before the pulse returns to it (the precise timing is set by the *TDG*), and the pulse now finds the output Pockels cell acting as a ¼ waveplate. It is rotated 45° going in and 45° reflecting back from CM_4 and becomes vertically polarized. It is reflected out of the cavity by the horizontal polarizer.
3. The vertically polarized, amplified pulse is reflected by the polarizer (15) to mirror M_5 (16).
4. To protect the compressor optics, the beam is expanded by a telescope (17) (comprising negative and positive lenses OL_1 and OL_2) to reduce pulse power density.
5. The expanded beam is directed into the compressor by the polarization rotating periscope PS_2 (18), which changes the vertically polarized light from the amplifier to horizontally polarized light.
6. M6 (see Figure 7-1) directs the beam into the compressor chamber.

The Pump Beam Path

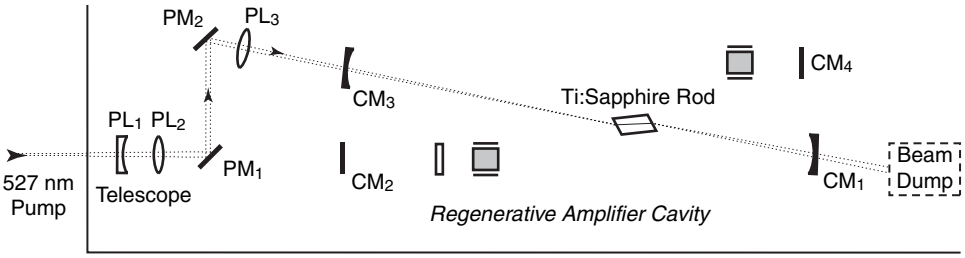


Figure 7-9: Pump Beam Path

The path of the pump beam is controlled as follows:

A telescope comprising negative lens PL_1 and positive lens PL_2 enlarge the pulsed beam from the pump laser. (The beam is enlarged so that it can be better focused into the Ti:sapphire rod.) The PL_2 lens mount has a longitudinal adjustment.

Pump mirror PM_1 directs the enlarged beam from the telescope to PM_2. PM_1 and PM_2 are used to set the height of the pump beam so that it is centered on pump lens PL_3. PL_3 focuses the pump beam in the Ti:sapphire rod, after the beam first passes through cavity mirror CM_3, which is coated to allow pas-

sage of the green pump light while reflecting the infrared pulses in the amplifier.

Pump mirrors PM_1 and PL_3 have vertical and horizontal adjustments. The PL_2 lens mount has a longitudinal adjustment (i.e., it moves the lens along the direction of the beam).

The 527 nm (green) light from the Nd:YLF pump laser is horizontally polarized, allowing it to enter the Ti:sapphire rod where roughly 80% of it is absorbed by the rod.

The fraction of the high-power pump beam that is not absorbed by the Ti:sapphire rod transmits through the rod onto CM_1. CM_1 does not reflect a significant amount of 527 nm light; instead, the coating for CM_1 allows the green beam to pass through the mirror, where it is absorbed by the beam dump behind the mirror.

Chapter 8 Maintenance and Troubleshooting



Exceptional care must be taken when operating the *Spitfire Pro* with the covers removed. Laser protective eyewear *must* be worn to protect the eyes from all wavelength emissions.

Try This First

If the *Spitfire Pro* is producing pulses but performance has degraded, first verify the following:

- the BWD interlock is properly set
- the seed laser is operating properly and its shutter is open
- the pump laser is operating properly and its shutter is open
- the pump beam routing mirror PM_2 is properly aligned

If the components above are operating properly, you may only need to adjust the pump power or pump beam routing mirror PM_2 to optimize output. Refer to Appendix A, "Alignment," before attempting these adjustments.

If the *Spitfire Pro* is not producing amplified pulses, first verify the following:

- that there is sufficient pump power
- that the pump beam has not become misaligned
- that the PC2 delay for the *TDG* is sufficiently greater than the PC1 delay
- that any safety switch connected to the INTERLOCK switch on the *TDG* back panel has not been activated
- that the Pockels cells are properly connected, triggered and operating
- that the intracavity apertures are not blocking the beam

If these criteria are all met, inspect the internal optics for cleanliness and damage. Use the procedure below to clean optics as needed. *Heed the warnings regarding cleaning!* Not all optical surfaces can be cleaned, except for blowing dust off with dry nitrogen.

If the troubleshooting and corrective procedures in this chapter do not solve the problem, please contact your Spectra-Physics representative before taking further action. Contact information is included in "Customer Service" on page 8-6.

Cleaning Optics

The *Spitfire Pro* has been designed for minimal maintenance. However, from time to time, depending on the laboratory environment, it may be necessary to clean the optics. The following materials are required:

- Reagent-grade methanol or acetone
- Lens tissues
- Hemostat (surgical pliers)
- Eyedropper

Warning! 

Do not attempt to clean the surfaces of the gratings or the gold-coated mirrors. These optical surfaces can only be blown clean with dry nitrogen. Attempting to clean these components will permanently damage them.

The optics in the *Spitfire Pro* should be carefully cleaned with soft optical tissue and reagent-grade methanol or acetone as described below.

1. Always wash your hands first.
2. Wear finger cots whenever optics are handled.
3. Hold one sheet of lens tissue over the optic to be cleaned.
4. Using the eyedropper, place a single drop of reagent-grade methanol on top of the lens tissue.
5. Drag the lens tissue across the optic only once.
6. If a residue of solvent is left on the optic, repeat the procedure using less solvent and a new lens tissue until no residue remains.

For hard to reach optics:

1. Wear finger cots or gloves.
2. Fold a piece of lens tissue repeatedly to form a pad of approximately 1 cm wide.
3. Hold the pad with a hemostat so that about 3 mm of the folded edge protrudes from the hemostat blades.
4. Saturate the pad with methanol or acetone and shake dry.
5. Reach slightly on one edge of mirrors and wipe the surface of the mirrors toward the outside *in one motion*. *Use each pad only once!* Be very careful that the tip of the hemostats does not scratch the mirror.

Troubleshooting

Symptom: No Spitfire Pro output

Possible Cause:	Corrective Action:
BWD interlock is open	Check the two BWD indicators on the Main menu. If both indicators are on, reset the interlock button of the BWD. If one or both indicators are off, verify the seed laser is mode-locked and that the wavelength is centered as specified. Reset the BWD interlock button after restoring seed laser operation.
Seed laser is not functioning correctly	Refer to seed laser user's manual for further instructions.
Pump laser is not functioning correctly	Refer to the pump laser user's manual for further instructions.
Seed laser beam is misaligned	Optimize the seed laser alignment.
TDG controls are disabled	Verify the unit is turned on and that the delay settings for PC1, PC2 and SYNC are properly set.
Problem with high speed driver(s)	Contact your Spectra-Physics representative

Symptom: Regenerative amplifier power is below specification

Possible Cause:	Corrective Action:
Optics are dusty	Use dry nitrogen to blow dust from the optics, with particular attention to the pump path and regenerative amplifier optics.
Optics are damaged	Check the optical components in the regenerative amplifier. If an optic has been damaged, contact your Spectra-Physics representative to arrange to have the optic changed. It may be possible to use an undamaged portion of the optic face and realign the regenerative amplifier as a temporary solution.
Seed laser beam is misaligned	Optimize the seed laser beam alignment.
Pump laser power is low	Optimize pump laser power according to its user's manual.
Pump laser beam is misaligned	Optimize the alignment of the pump laser beam.
Regenerative Amplifier is misaligned	Refer to Appendix A, "Alignment".
Timing of Pockels cells is incorrect	Check the delay settings for PC1 and PC2 for the TDG on the Main menu.

Symptom: Pulse has broadened out of specification

Possible Cause:	Corrective Action:
Compressor delay may not be optimized	Adjust the compressor motor controller to get the shortest pulse.
Seed laser pulses are broadened	Check the seed laser bandwidth and center wavelength.
Pump laser power is low or unstable	See troubleshooting guide in the pump laser user's manual.
Stretcher misalignment is broadening pulses	Check the alignment of the stretcher. Verify the seed laser beam alignment is optimized.
Pockels cells timing is incorrect	Check the delay settings for PC1 and PC2 for the TDG on the Main menu.
Optical components are damaged	Contact your Spectra-Physics representative.
"Wings" are present on output auto-correlation	Check the seed laser bandwidth and center wavelength. Verify the stretcher mirror M_2 is at the focal point of the large gold mirror, M_1. Verify the compressor grating is parallel to the stretcher grating (not applicable to the *Spitfire Pro 40F*). Contact your Spectra-Physics representative if this is not the case. Verify the beam is not clipped by the internal apertures.

Symptom: Output power or output spectrum is unstable

Possible Cause:	Corrective Action:
Power variation in the pump laser	See troubleshooting guide in the pump laser *User's Manual*.
Power variation in the regenerative amplifier	Check the delay settings for PC1 and PC2 for the TDG on the spitfire GUI Main display. Check chiller flow and water level.
Spectrum is modulated:	
Incorrect adjustment of the ¼ wave voltage to one or both Pockels cells	Contact your Spectra-Physics representative.
Excessive jitter on Spitfire Pro output pulse:	
Unstable seed laser performance	See troubleshooting guide in the seed laser user's manual.
Incorrect timing of the INPUT POCKELS CELL	Check the delay settings for PC1 on the Main menu.
Defective TDG or failure of the high speed driver(s)	Contact your Spectra-Physics representative.

Symptom: Poor contrast ratio

Possible Cause:	Corrective Action:
Pre-pulse:	
Incorrect alignment of the output Pockels cell	Contact your Spectra-Physics representative
Incorrect alignment of the ¼ wave plate for the input Pockels cell	Contact your Spectra-Physics representative
Post-pulse:	
Incorrect alignment of the Input Pockels cell	Adjust timing of the Output Pockels cell
Incorrect alignment of the Input Pockels cell	Contact your Spectra-Physics representative
Incorrect adjustment of the ¼ wave voltage for Output Pockels cell	Contact your Spectra-Physics representative

Symptom: Poor output beam quality

Possible Cause:	Corrective Action:
Incorrect pump beam alignment	Refer to Appendix A for pump beam alignment procedures.
Damage to optical components	Check for optical damage. Contact your Spectra-Physics representative if damage is present.
Compressor vertical retro-reflector and/or horizontal retro-reflector are incorrectly aligned in the horizontal axis	Contact your Spectra-Physics representative.

Symptom: Optical damage in the amplifier cavity

Possible Cause:	Corrective Action:
Seed laser is not well modelocked (CW breakthrough is present)	Contact your Spectra-Physics representative.
Partial restriction of the stretched spectrum	Contact your Spectra-Physics representative.
Failure to remove alignment tools from the optical path after checking stretcher or compressor alignment	Contact your Spectra-Physics representative.

Customer Service

At Spectra-Physics, we take great pride in the reliability of our products. Considerable emphasis has been placed on controlled manufacturing methods and quality control throughout the manufacturing process. Nevertheless, even the finest precision instruments will need occasional service. We feel our instruments have excellent service records compared to competitive products, and we hope to demonstrate, in the long run, that we provide excellent service to our customers in two ways: first by providing the best equipment for the money, and second, by offering service facilities that get your instrument repaired and back to you as soon as possible.

Spectra-Physics maintains major service centers in the United States, Europe, and Japan. Additionally, there are field service offices in major United States cities. When calling for service inside the United States, dial our toll free number: **1 (800) 456-2552**. To phone for service in other countries, refer to the section "Service Centers" on page 8-8.

Order replacement parts directly from Spectra-Physics. For ordering or shipping instructions, or for assistance of any kind, contact your nearest sales office or service center. You will need your instrument model and serial numbers available when you call. Service data or shipping instructions will be promptly supplied.

To order optional items or other system components, or for general sales assistance, dial **1 (800) SPL-LASER** in the United States, or **1 (650) 961-2550** from anywhere else.

Warranty

This warranty supplements the warranty contained in the specific sales order. In the event of a conflict between documents, the terms and conditions of the sales order shall prevail.

Unless otherwise specified, all parts and assemblies manufactured by Spectra-Physics are unconditionally warranted to be free of defects in workmanship and materials for a period of **one year** following delivery of the equipment to the F.O.B. point.

Liability under this warranty is limited to repairing, replacing or giving credit for the purchase price of any equipment that proves defective during the warranty period, provided prior authorization for such return has been given by an authorized representative of Spectra-Physics. Spectra-Physics will provide at its expense all parts and labor and one-way return shipping of the defective part or instrument (if required). In-warranty repaired or replaced equipment is warranted only for the remaining portion of the original warranty period applicable to the repaired or replaced equipment.

This warranty does not apply to any instrument or component not manufactured by Spectra-Physics. When products manufactured by others are included in Spectra-Physics equipment, the original manufacturer's warranty is extended to Spectra-Physics customers.

When products manufactured by others are used in conjunction with Spectra-Physics equipment, this warranty is extended only to the equipment manufactured by Spectra-Physics.

This warranty also does not apply to equipment or components that, upon inspection by Spectra-Physics, discloses to be defective or unworkable due to abuse, mishandling, misuse, alteration, negligence, improper installation, unauthorized modification, damage in transit, or other causes beyond the control of Spectra-Physics.

This warranty is in lieu of all other warranties, expressed or implied, and does not cover incidental or consequential loss.

The above warranty is valid for units purchased and used in the United States only. Products shipped outside the United States are subject to a warranty surcharge.

Return of the Instrument for Repair

Contact your nearest Spectra-Physics field sales office, service center, or local distributor for shipping instructions or an on-site service appointment. You are responsible for one-way shipment of the defective part or instrument to Spectra-Physics.

We encourage you to use the original packing boxes to secure instruments during shipment. If shipping boxes have been lost or destroyed, we recommend that you order new ones. We will return instruments only in Spectra-Physics containers.

Warning! 

Always drain and **fully vacuum-evacuate** the cooling water from the laser head and chiller before shipping. Water expands as it freezes and will damage the laser. Even during warm spells or summer months, freezing may occur at high altitudes or in the cargo hold of aircraft. Such damage is excluded from warranty coverage.

Service Centers

Belgium

Telephone: (32) 0800 1 12 57

France

Telephone: (33) 0810 00 76 15

Germany and Export Countries*

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**And all European and Middle Eastern countries not included on this list.*
***And all non-European or Middle Eastern countries not included on this list.*

Appendix A Alignment

Before starting these procedures, it is essential that you read Chapter 2, "Laser Safety," and that you become thoroughly familiar with the components and optical design of the *Spitfire Pro* as discussed in Chapter 3.



Use of controls or adjustments, or performance of procedures other than those specified herein may result in hazardous radiation exposure.



The following procedures are not intended for the initial installation of the *Spitfire Pro* amplifier. Call your Spectra-Physics service representative to arrange an installation appointment, which is part of your purchase agreement. *Allow only authorized Spectra-Physics representatives to install your laser.* You will be charged for repair of any damage incurred if you attempt to install the system yourself, and such action may void your warranty.

These procedures are supplied as a convenience in the event your *Spitfire Pro* system is out of warranty and a service call is problematic. These procedures might well result in loss of function or even permanent damage to the system if performed by personnel not trained by Spectra-Physics.

If the amplifier is no longer lasing, or if an intracavity optical component is seriously misaligned or damaged and must be replaced, contact your Spectra-Physics representative before attempting any repair. Experienced experts may be able to apprise you of techniques that might save you considerable time and expense in these circumstances.

More advanced procedures, such as replacing a damaged Ti:sapphire rod in the amplifier, will require a service call.

Essential to proper *Spitfire Pro* amplifier operation is the alignment of the seed laser into the amplifier, amplifier optimization and similar adjustments. These are described in Chapter 6, "Operation."

If the amplifier is operating properly with the installed optics set, but operation at a different wavelength range is required, contact your Spectra-Physics representative.

If you wish to convert the *Spitfire Pro* from femtosecond (*Spitfire Pro F)* to picosecond operation (*Spitfire Pro PM*), or the reverse, contact your Spectra-Physics representative. It is not necessary to realign the amplifier cavity for these procedures.

To understand these procedures, it is important to realize the *Spitfire Pro* can operate as a laser as well as an amplifier— that is, it can be configured to produce its own pulsed output (when energized by the pump laser) even when the input from the seed laser is blocked.

Begin these procedures with the pump laser and the seed laser on and warmed up, with both beams blocked (shuttered) from entering the *Spitfire Pro*.

Try This First

If your system is lasing but performance has degraded, only slight adjustments might be required in pump beam power or to the pump beam routing mirror, PM_2 (refer to Figure A-3) to optimize output, rather than performing a complete realignment.

Before beginning any realignment, verify the following:

- that there is sufficient pump power
- that the pump beam is properly aligned
- that the PC2 delay is sufficiently beyond PC1 (see "Basic Performance Optimization" on page 6-3.
- that any safety switch connected to the INTERLOCK switch on the *TDG* back panel has not been activated (i.e., that it is closed).
- that the Pockels cells are properly connected, triggered, and operating
- that the intracavity apertures are not blocking the beam

Tools Required:

- a power meter capable of measuring between 10 mW and 20 W of average power from 500 to 900 nm
- a fast CRT analog oscilloscope capable of 300 MHz or better
- a fast photodiode with a 2 ns rise time or better
- IR viewer and IR card
- a small, low-divergence HeNe laser (for alignment)
- an autocorrelator (e.g., Spectra-Physics PulseScout)
- scales (rulers) to measure up to 10 in. and 25 cm
- three gimbal mounts with 4–6 in. adjustable height
- three silver mirrors for the above mounts
- alignment pins
- #1 Phillips screwdriver
- a standard (English) hex ball driver set
- a metric hex driver set
- white business card
- reagent-grade methanol
- lens tissue
- gel linear polarizing film

Stretcher Alignment Check

If you cannot see the beam in the amplifier, it will be necessary to check the alignment through the stretcher, as follows:

1. Use an IR viewer to look at the beam pattern on the stretcher grating. It should look like that in Figure A-1.

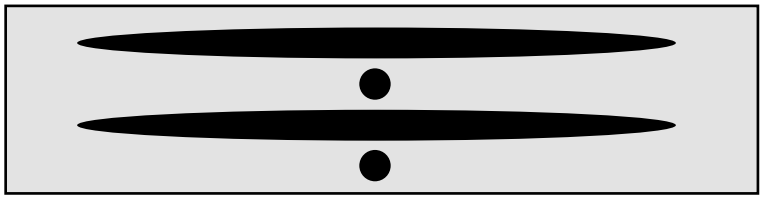


Figure A-1: Radiation Patterns on Stretcher Gratings

2. If the beam pattern does not look like Figure A-1, it is likely that the wavelength of the seed laser has changed. To return to the previous operating conditions, adjust the seed laser wavelength until the pattern is symmetrical on the grating as shown here.

   After adjusting the seed laser, re-check the beam pattern. If the pattern shown in Figure A-1 cannot be obtained, the stretcher may need to be realigned. This procedure is beyond the scope of this manual. Contact Spectra-Physics for assistance.

3. The output beam from the stretcher should now be picked off by mirror M_3. It may be necessary to *slightly* adjust the vertical tilt of the large gold mirror M_1 using the adjustment described in Chapter 7, "The *Spitfire Pro* Beam Path: Stretcher and Compressor Beam Paths."

4. Re-check the alignment of the beam into the regenerative amplifier.

Compressor Alignment Check

The alignment of the beam through the compressor should not have changed since the *Spitfire Pro* was last operated, but check it anyway.

Verify the output beam is round and even in intensity.

Use an IR viewer to look at the compressor grating. The pattern shown in Figure A-2 should be evident.



Figure A-2: Radiation Patterns on Compressor Gratings

Pump Beam Alignment

Once the *Spitfire Pro* system has been properly installed, the alignment of the pump laser into the *Spitfire Pro* amplifier should not need to be adjusted during normal operation. Some circumstances however, such as maintenance or service of the pump laser, may require aligning the pump beam. The most common symptom of pump beam misalignment is poor *Spitfire Pro* mode quality that results when there is poor superposition of the pump beam mode in the Ti:sapphire rod.

Refer to Chapter 7 for a description of the optical design shown in Figure A-3.

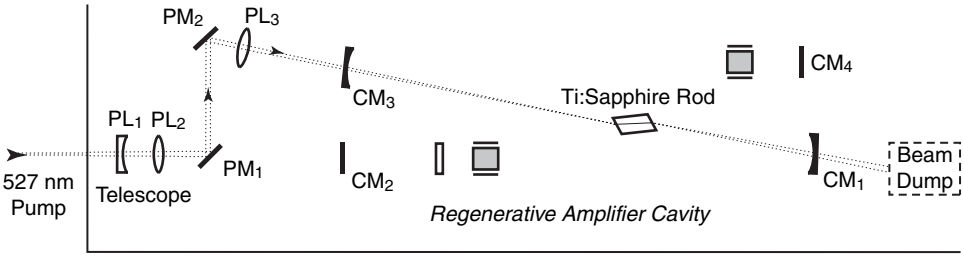


Figure A-3: Pump Beam Path of the *Spitfire Pro*

1. Record the pump beam power required to operate the *Spitfire Pro*.
2. Block the seed pulses into the *Spitfire Pro*.
3. Adjust the pump laser (*Empower*) power to the minimum power that still allows a stable green beam to be observed.
4. Verify the pump beam passes through the input port without clipping.
5. Adjust the pump beam to center it on the lenses of the zoom telescope (PL_1 and PL_2) and also on PM_1.
6. Adjust PM_1 to center the beam on PM_2.
7. Adjust PM_2 to center the beam in the Ti:sapphire rod.
8. The beam should be near the center of the back of mirror CM_3. If this is not the case, make small adjustments to PM_1 as needed.
9. When the pump beam is centered on the Ti:sapphire rod, it should pass through the center of PL_3. If it does not, loosen the screw holding the PL_3 mount to the chassis and position it so that it does. Be sure to maintain the distance from the lens to the face of the Ti:sapphire rod. If moving PL_3 moves the pump beam on the rod, use PM_2 to re-center it. Iterate adjustment of PM_2 and PL_3 until the beam is centered on both PL_3 and the rod.

   Note that having the correct distance from PL_3 to the Ti:sapphire rod is particularly important for maintaining proper pump beam mode in the rod. The correct placement of PL_3 should be marked on the base plate of the amplifier head assembly.
10. Return the pump laser to Q-switched operation, and adjust it to the power level noted in Step 1. If the required pump power is not known, set the pump power to 10 W.

11. The *Spitfire Pro* should now operate as a laser. If it does not, disable the output pockels cell by clicking on the button below PC2 on the Main menu (it turns gray) and scan the pump beam waist across the Ti:sapphire crystal face until it is superposed onto the intracavity beam waist. To do this, position the pump beam high in the crystal and off to one side using PM_2. Now slowly translate the pump beam across the crystal face to the opposite side. Lower the beam position in the crystal ~0.5 mm and slowly translate back across the crystal face. Continue this scanning process until the amplifier resonator begins to lase.

Note



The Ti:sapphire crystal emits *less* fluorescence when the *Spitfire Pro* begins to lase. While scanning, watch for this, rather than for an output from the thin-film polarizer, which necessitates watching *both* the crystal (for safety) *and* the output (for lasing).

12. Once the amplifier has begun to lase, position a power meter in front of the second lens of the beam expanding telescope, OL_2.
13. Enable the output pockels cell by again clicking on the button below PC2 (it turns green) and adjust its timing (turn the PC2 knob) so that cavity dumping occurs at saturation.
14. Adjust PM_2 vertically and horizontally for a symmetrically shaped output mode (it should approach a single-order mode). This should also coincide with maximum output power.
15. Verify the cavity beams are parallel to the chassis top surface (measure the mirror *leakage* beam height from the chassis surface *outside* the cavity behind CM_3 and CM_4) and correct any error.

    It is likely that the beam height beyond CM_4 is either too high or too low. To correct any error, make vertical adjustments to CM_3 while compensating for power and mode shape with vertical adjustments to CM_4 until the correct beam height is restored.



Laser radiation is present. Beware of the eye hazard from the residual pump beam behind all intracavity mirrors!

16. Repeat Step 15 for the beam between CM_2 and CM_1. Adjust CM_1 for proper beam height behind CM_2 and compensate for output power and mode shape with CM_2.
17. Reposition both intracavity apertures coaxially about the beam as follows. Open each aperture fully, then reduce the diameter of each iris. The beam should be symmetrical around the iris as it is reduced, passing through its center. If it is not, loosen the screw that retains the aperture post and center the iris about the 800 nm intracavity beam.
18. Open both intracavity apertures and record the output power.

The pump laser beam should now be optimally aligned.

Compressor Alignment

Danger!
Laser Radiation

In the following procedure, the regenerative amplifier is initially operated as an optically pumped, Q-switched laser. In this configuration, the *Spitfire Pro* is capable of producing >1.5 W of average power at a wavelength near 800 nm. Use appropriate caution.

This procedure assumes that the grating block assembly is properly aligned, as are the input beams into the stretcher and compressor, and that the *Spitfire Pro* is fully operational. Under these circumstances, the compressor requires only minimal adjustment for optimal performance.

1. Disable PC1 and PC2 by clicking on their Main menu buttons.
2. Close the shutters of the pump laser and the seed laser.

Warning!

Block the seed beam as called for in this procedure. Blocking the seed beam incorrectly will result in significant damage to amplifier components. Such damage is not covered by your warranty.

3. Remove the covers from the *Spitfire Pro*.
4. Prevent the stretched seed beam from entering the regenerative amplifier by placing a beam block after mirror M_4 as shown in Figure A-4.
5. Operate the *Spitfire Pro* as a Q-switched, cavity-dumped laser. To do this, enable PC1 and PC2 and adjust PC2 as necessary (see "Optimizing the Regenerative Amplifier" in Chapter 6).
6. Install the removable reference iris in the X_1 location (see Figure A-4), and adjust the aperture opening to about 4 mm diameter.

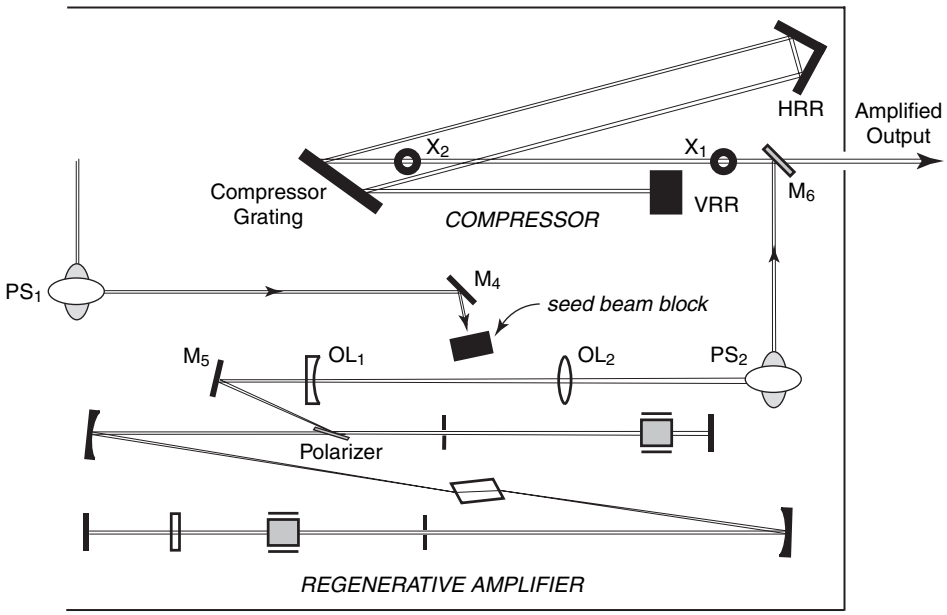


Figure A-4: Alignment of Amplifier Beam into the Compressor

7. Adjust mirror M_5 to center the cavity-dumped beam through the aperture of the removable iris.
8. Relocate the removable iris to new location X_2 shown in Figure A-4.
9. Adjust mirror M_6 to center the cavity-dumped beam through the aperture of the removable iris.
10. Return the aperture to the X_1 location and repeat Step 6. Iterate Steps 6 through 9 until the beam is aligned through the aperture in both locations with no further adjustments needed to either M_5 or M_6.
11. Remove the beam block installed in Step 4.
12. Check the buildup time reduction—it should be the same as when the seed beam optimization alignment procedure is performed (refer to "Seed Beam Alignment into the Regenerative Amplifier" on page 6-4).
13. Adjust the PC2 trigger timing for cavity dumping the correct pulse.
14. Verify that the cavity-dumped output is centered on the iris at location X_2.
15. Using the IR viewer to look at the grating, verify the pattern on the grating appears the same as that in Figure A-2. If not, go back and check the alignment through the iris.
16. Remove the iris from the *Spitfire Pro*, then verify the output beam is not clipped.
17. The compressor alignment should now be optimized. If this is not the case, contact your Spectra-Physics service engineer.

Ejecting the Pulse from the Amplifier

After performing the alignment procedures above, it will be necessary to adjust the timing of the Pockels cells to achieve proper capture and ejection of pulses from the amplifier. Refer to Chapter 6 for this procedure.

Notes

Report Form for Problems and Solutions

We have provided this form to encourage you to tell us about any difficulties you have experienced in using your Spectra-Physics instrument or its manual—problems that did not require a formal call or letter to our service department, but that you feel should be remedied. We are always interested in improving our products and manuals, and we appreciate all suggestions.

Thank you.

From:

Name _____

Company or Institution _____

Department _____

Address _____

_____

Instrument Model Number _____ Serial Number _____

Problem: _____

_____

_____

_____

_____

Suggested Solution(s): _____

_____

_____

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